

Parabolic subgroups of reductive groups

SGA 3, Exposé XXVI

English draft

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About this draft

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Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Exposé XXVI. Parabolic subgroups of reductive groups

⁽⁰⁾ This exposé studies the parabolic subgroups of a reductive S -group G . Its essential result is the conjugacy theorem (5.4). The essential tool is the notion of transversal position of two parabolic subgroups, a notion studied systematically in no. 4. Another fact plays an important role: the decomposition of the unipotent radical $\text{rad}^u(P)$ of a parabolic subgroup P into successive extensions of vector groups (2.1).⁽¹⁾

Various schemes associated with G are studied in no. 3; no. 6 deals with the split subtori⁽²⁾ of G and their relations with the parabolic subgroups.

Finally, in no. 7, we briefly explain how, over a semilocal base, one formulates the “relative theory” of reductive groups as it is set out in the case of fields in the article by A. Borel and J. Tits, *Groupes réductifs*, Publications Mathématiques de l’IHÉS, no. 27. In this article, cited as [BT65] below, the reader will moreover find, in the case of a base field, other results which have not been touched upon here.

1. Recollections. Levi subgroups

Definition 1.1. Let S be a scheme, G a reductive S -group, and P a group S -subscheme of G . One says that P is a *parabolic subgroup* of G if

- (i) P is smooth over S ,
- (ii) for each $s \in S$, the quotient $\bar{s}$ -scheme $G_{\bar{s}}/P_{\bar{s}}$ is proper (i.e. Bible, § 6.4, th. 4 (= [Ch05], § 6.5, th. 5), $P_{\bar{s}}$ contains a Borel subgroup of $G_{\bar{s}}$).

Proposition 1.2 (Exp. XXII, 5.8.5). *Let S be a scheme, G a reductive S -group, and P a parabolic subgroup of G . Then P is closed in G , with connected fibers, and*

$$P = \underline{\text{Norm}}_G(P).$$

Moreover, the quotient sheaf G/P is representable by an S -scheme smooth and projective over S .

Proposition 1.3 (Exp. XXII, 5.3.9 and 5.3.11). *Let S be a scheme, G a reductive S -group, and P and P' two parabolic subgroups of G . The following conditions are equivalent:*

- (i) P and P' are conjugate in G , locally for the étale (resp. (fpqc)) topology.
- (ii) For each $s \in S$, $P_{\bar{s}}$ and $P'_{\bar{s}}$ are conjugate by an element of $G(\bar{s})$.
- (iii) The strict transporter $\underline{\text{Transt}}_G(P, P')$ of P into P' (defined by

$$\underline{\text{Transt}}_G(P, P')(S') = \{g \in G(S') \mid \text{int}(g)P_{S'} = P'_{S'}\}$$

for every $S' \rightarrow S$) is a closed subscheme of G , smooth and of finite presentation over S , which is a principal homogeneous bundle on the right under P , and on the left under P' .

⁽⁰⁾**N.D.E.** Version of 13/10/2024.

⁽¹⁾**N.D.E.** Since the Levi subgroups of P form a torsor under $\text{rad}^u(P)$ (1.9), this implies, when S is semilocal, that P possesses a Levi subgroup and hence a maximal torus (2.4).

⁽²⁾**N.D.E.** The terminology “trivial torus” has been replaced by “split torus”.

Proposition 1.4. *Let S be a nonempty scheme, (G, T, M, R) a split reductive S -group, and R' a subset of R . The following conditions on R' are equivalent:*

- (i) $\mathfrak{g}_{R'} = \mathfrak{t} \oplus \coprod_{\alpha \in R'} \mathfrak{g}^\alpha$ is the Lie algebra of a parabolic subgroup of G containing T (necessarily unique, Exp. XXII, 5.3.5).
- (ii) R' is of type (R) (Exp. XXII, 5.4.2) and contains a system of positive roots.
- (iii) R' is a closed subset of R and satisfies: if $\alpha \in R - R'$, then $-\alpha \in R'$ (that is, $R = R' \cup (-R')$).
- (iv) There exist a system of simple roots Δ and a subset A of Δ such that R' is the union of the set of positive roots and the set of negative roots which are linear combinations of the elements of A .
- (v) R' contains a system of simple roots of R ; moreover, if $\Delta \subset R'$ is a system of simple roots of R , and if one puts

$$A = (-R') \cap \Delta,$$

then R' is the union of the set of positive roots and the set of negative roots which are linear combinations of the elements of A .

One has (i) $\Leftrightarrow$ (ii) by Exp. XXII, 5.4.5 (ii) and 5.5.1. One has (iii) $\Rightarrow$ (ii) by Exp. XXI, 3.3.6 and Exp. XXII, 5.4.7. One evidently has (v) $\Rightarrow$ (iv) $\Rightarrow$ (iii). One has (iii) $\Rightarrow$ (v) by Exp. XXI, 3.3.6 and 3.3.10.

It therefore remains to prove that (i) implies that R' is a closed subset of R . Now this last assertion can be checked on any geometric fiber; one can therefore suppose that S is the spectrum of an algebraically closed field.

Let P be the parabolic subgroup of G containing T whose Lie algebra is $\mathfrak{g}_{R'}$. Since the Borel subgroups of P are the Borel subgroups of G contained in P , it follows from Bible, § 12.3, th. 1, and from Exp. XXII, 5.4.5 (i) that, if one denotes by U the unipotent radical of P , then $T \cdot U$ is the subgroup of G containing T with Lie algebra

$$\mathfrak{g}_{R''} = \mathfrak{t} \oplus \coprod_{\alpha \in R''} \mathfrak{g}^\alpha,$$

where R'' is the intersection of the systems of positive roots of R contained in R' .

It follows in particular that R'' is closed and that $R'' \cap (-R'') = \emptyset$. On the other hand, the group $H = P/U$ is reductive, the canonical image $\overline{T}$ of T is a maximal torus of it ($T \rightarrow \overline{T}$ is an isomorphism), and one has an isomorphism of T -modules, i.e. of vector spaces graded of type M ,

$$\mathrm{Lie}(H) \simeq \mathfrak{t} \oplus \coprod_{\alpha \in R_s} \mathfrak{g}^\alpha,$$

where R_s is the complement of R'' in R' . It follows that R_s is naturally identified with the set of roots of H relative to $\overline{T}$, and in particular satisfies $R_s = -R_s$. It follows immediately that one has

$$R'' = \{\alpha \in R' \mid -\alpha \notin R'\}, \quad R_s = \{\alpha \in R' \mid -\alpha \in R'\}.$$

Let us now show that R' is closed. Let $\alpha, \beta \in R'$ be such that $\alpha + \beta \in R$; let us prove that $\alpha + \beta \in R'$. If $\alpha, \beta \in R''$, then $\alpha + \beta \in R''$, since R'' is closed. If $\alpha \in R_s$, $\beta \in R''$, and if $\alpha + \beta \notin R'$, then $\alpha + \beta \in -R''$, and one has $-\alpha = -(\alpha + \beta) + \beta \in R''$, since R'' is closed, which implies $-\alpha \in R'' \cap R_s$ and contradicts the fact that $R_s \cap R'' = \emptyset$. It therefore remains to study the case where $\alpha, \beta \in R_s$. If $\alpha + \beta \notin R'$, then $\alpha + \beta \in -R''$. But since $\alpha + \beta \neq 0$, there exists a system of positive roots of the root system R_s containing α and β , hence a Borel subgroup of $H = P/U$ containing the canonical images of U_α and U_β . Its inverse image in P is a Borel subgroup containing U_α , U_β and U , hence U_α , U_β and $U_{-(\alpha+\beta)}$, which is impossible.

Corollary 1.5. *A parabolic subgroup of a reductive group is of type (RC) (Exp. XXII, 5.11.1).*

Proposition 1.6 (Exp. XXII, 5.11.4). *Let S be a scheme, G a reductive S -group, and P a subgroup of type $(RC)^{(3)}$ of G .*

- (i) *P possesses a largest invariant group subscheme, smooth and of finite presentation over S , with connected unipotent geometric fibers. This is a characteristic subgroup of P , called the unipotent radical of P , denoted $\text{rad}^u(P)$. The quotient sheaf $P/\text{rad}^u(P)$ is representable by a reductive S -group.*
- (ii) *If T is a maximal torus of P , P possesses a reductive subgroup L containing T such that:*
 - (a) *Every reductive subgroup of P containing T is contained in L .*
 - (b) *P is the semidirect product $L \cdot \text{rad}^u(P)$, i.e. the canonical morphism $L \rightarrow P/\text{rad}^u(P)$ is an isomorphism.*

Moreover, L is the unique subgroup (resp. reductive subgroup) of P containing T and satisfying (b) (resp. (a)). Finally, one has

$$\underline{\text{Norm}}_P(L) = L, \quad \underline{\text{Norm}}_P(T) = \underline{\text{Norm}}_L(T).$$

1.7. — A subgroup L of P satisfying condition (b) above is called a *Levi subgroup* of P . It is a maximal reductive subgroup of P ; indeed, it is reductive, since it is isomorphic to $P/\text{rad}^u(P)$; let us show that it is maximal for this property. Let L' be a reductive subgroup of P containing L ; to prove that $L' = L$, one can reason locally for the (fpqc) topology, and hence suppose that L possesses a maximal torus T , and one is reduced to 1.6 (ii).

If L and L' are two Levi subgroups of P , L and L' are conjugate in P , locally for the (fpqc) topology. Indeed, locally for this topology, one can suppose that L (resp. L') possesses a maximal torus T (resp. T'); since T and T' are conjugate in P locally for the (fpqc) topology, one can suppose $T = T'$, and one then has $L = L'$ by 1.6 (ii). But since $P = L \cdot \text{rad}^u(P)$ and $\underline{\text{Norm}}_P(L) = L$, one immediately deduces:

Corollary 1.8. *Let P be a subgroup of type $(RC)^{(4)}$ of the reductive S -group G . If L and L' are two Levi subgroups of P , there exists a unique $u \in \text{rad}^u(P)(S)$ such that $\text{int}(u)L = L'$.*

Denote by $\underline{\text{Lev}}(P)$ the functor of Levi subgroups of P : for $S' \rightarrow S$, $\underline{\text{Lev}}(P)(S')$ is the set of Levi subgroups of $P_{S'}$. One deduces from 1.8:

Corollary 1.9. *Let P be a subgroup of type $(RC)^{(4)}$ of the reductive S -group G . Then $\underline{\text{Lev}}(P)$ is a principal homogeneous bundle under the S -group $\text{rad}^u(P)$, and in particular is representable by an S -scheme smooth and affine over S , with integral geometric fibers.*

It follows immediately from 1.6:

Corollary 1.10. *Let P be a subgroup of type $(RC)^{(4)}$ of the reductive S -group G . The functor $\underline{\text{Tor}}(P)$ of maximal tori of P is representable by a smooth affine S -scheme; the “relation $L \supset T$ ” defines a morphism*

$$\underline{\text{Tor}}(P) \rightarrow \underline{\text{Lev}}(P),$$

the fiber of this morphism above $L \in \underline{\text{Lev}}(P)(S)$ is identified with $\underline{\text{Tor}}(L)$ (Exp. XXII, 5.8.3).

The first assertion of 1.10 is a consequence of the other two and of Exp. XXII, 5.8.3.

Definition 1.11. Let S be a nonempty scheme, G a reductive S -group, P a parabolic subgroup of G , and $\mathcal{E} = (T, M, R, \Delta, (X_\alpha)_{\alpha \in \Delta})$ a pinning of G . One says that $\mathcal{E}$ is *adapted to P* , or that $\mathcal{E}$ is a *pinning of the pair (G, P)* , if $P \supset T$ and if the Lie algebra of P is of the form $\mathfrak{t} \oplus \coprod_{\alpha \in R'} \mathfrak{g}^\alpha$, where R' is a subset of R containing R_+ .

⁽³⁾**N.D.E.** “Parabolic subgroup” has been replaced by “subgroup of type (RC)”, as in Exp. XXII, 5.11.4, since it will be useful later (4.5.1, 6.17) to be able to apply this statement to $P \cap P'$, when P, P' are two parabolic subgroups such that $P \cap P'$ is of type (RC).

⁽⁴⁾**N.D.E.** Cf. N.D.E. (3).

In particular, if $T \subset B$ is the Killing pair defined by the pinning, one has $T \subset B \subset P$.

Under the preceding conditions, one puts $\Delta(P) = \Delta \cap (-R')$; then, by Exp. XXII, 5.4.3, one has:

$$\begin{aligned} \alpha \in \Delta(P) &\iff \alpha \in \Delta \text{ and } U_{-\alpha} \subset P \\ &\iff \alpha \in \Delta \text{ and } U_{-\alpha} \cap P \neq e. \end{aligned}$$

It follows immediately from 1.4 (v) and Exp. XXII, 5.11.3 and 5.10.6:

Proposition 1.12. *Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G , $(T, M, R, \Delta, (X_\alpha)_{\alpha \in \Delta})$ a pinning of G adapted to P , and $\Delta(P)$ the subset of Δ defined above.*

(i) *The unipotent radical $\text{rad}^u(P)$ of P is precisely*

$$U_{R''} = \prod_{\alpha \in R''} U_\alpha,$$

where R'' is the set of positive roots which, in their decomposition on Δ , contain at least one element of $\Delta - \Delta(P)$ ⁽⁵⁾ with nonzero coefficient.

(ii) *The unique Levi subgroup L of P containing T is precisely*

$$Z_{\Delta(P)} = \underline{\text{Centr}}_G(T_{\Delta(P)}),$$

where $T_{\Delta(P)}$ is the maximal torus of $\bigcap_{\alpha \in \Delta(P)} \text{Ker}(\alpha)$; moreover one has $T_{\Delta(P)} = \text{rad}(L)$.

Corollary 1.13. *Every Levi subgroup L of the parabolic subgroup P of the reductive group G is a critical subgroup of G , i.e. satisfies (Exp. XXII, 5.10.4):*

$$L = \underline{\text{Centr}}_G(\text{rad}(L)).$$

This follows immediately from 1.12 and the following lemma, contained in 1.4 and Exp. XXII, 5.4.1:

Lemma 1.14. *Locally for the étale topology, every pair (G, P) , where P is a parabolic subgroup of the reductive group G , can be pinned (1.11).*

Let us note:

Proposition 1.15. *Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G , and $\mathcal{E}$ and $\mathcal{E}'$ two pinnings of G adapted to P . The unique inner automorphism of G over S which transforms $\mathcal{E}$ into $\mathcal{E}'$ (Exp. XXIV, 1.5) comes from P , by the morphism*

$$P \longrightarrow P/\underline{\text{Centr}}(P) = P/\underline{\text{Centr}}(G) \longrightarrow G/\underline{\text{Centr}}(G).$$

Indeed, it suffices to reason as in Exp. XXIV, 1.5, using:

Lemma 1.16 (Exp. XXII, 5.3.14 and 5.2.6). *The maximal tori (resp. Borel subgroups, resp. Killing pairs) of a parabolic subgroup P of the reductive S -group G are conjugate in P , locally for the étale topology.*

Proposition 1.17. *Let S be a scheme, G a reductive S -group, P and P' two parabolic subgroups of G , and B a Borel subgroup contained in P and P' . If P and P' are conjugate in G , locally for the étale topology, then $P = P'$.*

Indeed, one can suppose that there exists $g \in G(S)$ such that $\text{int}(g)P = P'$. Then B and $\text{int}(g)^{-1}B$ are two Borel subgroups of P . Extending S , one can, by 1.16, suppose that there exists $p \in P(S)$ such that $\text{int}(p)\text{int}(g^{-1})B = B$. Then

$$pg^{-1} \in \underline{\text{Norm}}_G(B)(S) = B(S),$$

and $g \in B(S) \cdot p \subset P(S)$, hence $P' = \text{int}(g)P = P$.

⁽⁵⁾**N.D.E.** $\Delta(P)$ has been corrected to $\Delta - \Delta(P)$.

Remark 1.18. If P and P' are two parabolic subgroups of G containing the same Borel subgroup, then $P \cap P'$ is again a parabolic subgroup of G . Indeed, it is smooth along the identity section (Exp. XXII, 5.4.5), and it contains a Borel subgroup.

Proposition 1.19. *Let S be a scheme, G a reductive S -group, and G' its derived group (Exp. XXII, 6.2.1).*

(i) *The maps*

$$P \mapsto P' = P \cap G' \quad \text{and} \quad P \mapsto P' \cdot \text{rad}(G) = \underline{\text{Norm}}_G(P')$$

are mutually inverse bijections between the set of parabolic subgroups of G and the set of parabolic subgroups of G' . One has $\text{rad}^u(P) = \text{rad}^u(P')$.

(ii) *Let P be a parabolic subgroup of G , and $P' = P \cap G'$. The maps*

$$L \mapsto L' = L \cap G' = L \cap P' \\ L' \mapsto L' \cdot \text{rad}(G) = \underline{\text{Centr}}_G(\text{rad}(L'))$$

are mutually inverse bijections between the set of Levi subgroups of P and the set of Levi subgroups of P' . Moreover, one has $\text{rad}(L') = (\text{rad}(L) \cap G')^0$.

The proof (by reduction to the split case, for example) presents no difficulty and is left to the reader, as is the immediate proof of:

Proposition 1.20. *Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G , and L a Levi subgroup of P . The maps*

$$Q \mapsto Q \cap L = Q', \quad Q' \mapsto Q' \cdot \text{rad}^u(P)$$

are mutually inverse bijections between the set of parabolic subgroups of G contained in P and the set of parabolic subgroups of L . Moreover, the Levi subgroups of Q' are the Levi subgroups of Q contained in L .

One can supplement 1.6 as follows:

Proposition 1.21. *Let S be a scheme, G a reductive S -group, and P a parabolic subgroup of G .*

- (i) *P possesses a largest invariant subgroup, smooth and of finite presentation over S , with connected solvable geometric fibers. This is a characteristic subgroup of P , called the radical of P , and denoted $\text{rad}(P)$. The quotient sheaf $P/\text{rad}(P)$ is representable by a semisimple S -group.*
- (ii) *If L is a Levi subgroup of P , $\text{rad}(P)$ is the semidirect product of $\text{rad}^u(P)$ and $\text{rad}(L)$; one has $\text{rad}(L) = L \cap \text{rad}(P)$, hence $L = \underline{\text{Centr}}_G(L \cap \text{rad}(P))$, and $P/\text{rad}(P) \simeq L/\text{rad}(L)$.*

Indeed, since assertion (i) is local, one can suppose that G possesses a Levi subgroup L , and one is reduced to proving that $R = \text{rad}^u(P) \cdot \text{rad}(L)$ possesses the properties stated in (i), which is immediate. For (ii), it remains only to prove that $\text{rad}(L) = L \cap \text{rad}(P)$, which follows immediately from the fact that $L \cap \text{rad}(P)$ is smooth with connected fibers, L being the centralizer of a torus.⁽⁶⁾

2. Structure of the unipotent radical of a parabolic subgroup

Proposition 2.1. *Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G , and $\text{rad}^u(P)$ its unipotent radical. There exists a sequence of group subschemes of $\text{rad}^u(P)$*

$$\text{rad}^u(P) = U_0 \supset U_1 \supset U_2 \supset \cdots \supset U_n \supset \cdots$$

possessing the following properties:

⁽⁶⁾**N.D.E.** Cf. XIX 1.4.

- (i) Each U_i is smooth, with connected fibers, characteristic and closed in P . The commutator of a section of U_i and a section of U_j is a section of U_{i+j+1} (over a variable $S' \rightarrow S$).
- (ii) For each $i \geq 0$, there exist a locally free $\mathcal{O}_S$ -module $\mathcal{E}_i$ and an isomorphism of S -sheaves of groups

$$U_i/U_{i+1} \xrightarrow{\sim} W(\mathcal{E}_i).$$

- Moreover, the automorphisms of P (over a variable $S' \rightarrow S$) act linearly on $W(\mathcal{E}_i)$.
- (iii) For every $s \in S$, one has $U_{n,s} = e$ for $n > \dim(\text{rad}^u(P)_s)$.

2.1.1. — First suppose that the pair (G, P) is pinnable. Let $(T, M, R, \Delta, \dots)$ be a pinning of G adapted to P ; let $\Delta(P)$ be the subset of Δ defined by P . Let $\alpha_1, \dots, \alpha_p$ be the elements of $\Delta(P)$, and $\beta_1, \dots, \beta_q$ the elements of $\Delta - \Delta(P)$. Every root $\gamma \in R$ is written uniquely as

$$\gamma = a_1\alpha_1 + \dots + a_p\alpha_p + b_1\beta_1 + \dots + b_q\beta_q.$$

Put

$$a(\gamma) = b_1 + \dots + b_q. \quad (7)$$

The following properties follow immediately from the definitions (cf. 1.12):

- (i) $U_\gamma \subset P \Leftrightarrow a(\gamma) \geq 0$.
- (ii) $U_\gamma \subset \text{rad}^u(P) \Leftrightarrow a(\gamma) > 0$.
- (iii) $a(n\gamma + m\gamma') = na(\gamma) + ma(\gamma')$ for all $n, m \in \mathbf{Z}$.

For $i > 0$, let R_i be the set of roots $\gamma \in R$ such that $a(\gamma) > i$. Each R_i is a closed set of roots satisfying $R_i \cap (-R_i) = \emptyset$. Consider (Exp. XXII, 5.6.5) the S -group

$$U_i = U_{R_i} = \prod_{\gamma \in R_i} U_\gamma.$$

This is a closed group subscheme of G , smooth over S , with connected fibers.

Let $\alpha, \beta \in R$; consider the commutation relation of Exp. XXII, 5.5.2

$$p_\alpha(x)p_\beta(y)p_\alpha(-x) = p_\beta(y) \prod_{n,m \in \mathbf{N}^*} p_{n\alpha+m\beta}(C_{n,m,\alpha,\beta}x^n y^m),$$

where each p_γ is an isomorphism of vector groups $\mathbf{G}_{a,S} \xrightarrow{\sim} U_\gamma$. First note that, if $a(\alpha) > i$ and $a(\beta) > j$, one has

$$a(n\alpha + m\beta) = na(\alpha) + ma(\beta) \geq n(i+1) + m(j+1) > i+j+1$$

when n and m are > 0 . It follows that the commutator of a section of U_α and a section of U_β is a section of U_{i+j+1} (over a variable $S' \rightarrow S$), which indeed implies $(U_i, U_j) \subset U_{i+j+1}$. For each $i \geq 0$, the quotient U_i/U_{i+1} is therefore commutative; it is naturally identified with

$$U_i/U_{i+1} \simeq \prod_{a(\gamma)=i+1} U_\gamma \simeq W(\mathcal{E}_i),$$

where $\mathcal{E}_i$ is the direct sum of the $\mathfrak{g}^\gamma$ for $a(\gamma) = i+1$.

Let us return to the commutation formula above, and suppose $a(\alpha) \geq 0$, $a(\beta) > i$. If $n, m \in \mathbf{N}^*$,

- either $a(n\alpha + m\beta) > i+1$,
- or $a(n\alpha + m\beta) = i+1$, in which case one necessarily has $m = 1$.

This first proves that $\text{int}(p_\alpha(x))$ preserves U_i (hence also U_{i+1}), then that the expression $\text{int}(p_\alpha(x))p_\beta(y)$ involves modulo U_{i+1} only terms of the form $p_{n\alpha+\beta}(C_{n,1,\alpha,\beta}x^n y)$, which are therefore *linear* in y . It follows that the inner automorphisms defined by sections of U_α act linearly on the quotient U_i/U_{i+1} identified with $W(\mathcal{E}_i)$. Since this is also trivially true for the inner automorphisms defined by sections of T , and since P is generated by T and the U_α , $a(\alpha) \geq 0$, one deduces that:

- (i) each U_i is invariant in P ,

(7)**N.D.E.** $a_1 + \dots + a_p$ has been corrected to $b_1 + \dots + b_q$.

(ii) the inner automorphisms defined by sections of P act linearly on $U_i/U_{i+1} \simeq W(\mathcal{E}_i)$.

2.1.2. — Now let $(T', M', R', \Delta', \dots)$ be a new pinning of G adapted to P . By 1.15, there exists an inner automorphism of G coming from P which transforms the old pinning into the new one. Extending S , one can suppose that this inner automorphism is of the form $\text{int}(p)$, $p \in P(S)$.

If one repeats the preceding constructions using the new pinning, it is clear that the groups U'_i and the isomorphisms $U'_i/U'_{i+1} \simeq W(\mathcal{E}'_i)$ obtained are deduced from U_i and $U_i/U_{i+1} \simeq W(\mathcal{E}_i)$ by transport of structure by means of $\text{int}(p)$. It follows from remarks (i) and (ii) above that one will therefore have $U'_i = U_i$, and that the two vector structures constructed on $U_i/U_{i+1} = U'_i/U'_{i+1}$ coincide.

This shows us that the groups U_i and the vector structures on the quotients U_i/U_{i+1} are independent of the pinning considered (and in particular invariant under every automorphism of P , as one easily sees).

We have therefore proved the proposition when the pair (G, P) is pinnable (part (iii) is trivial, since by Exp. XXI 3.1.2, the set $\{a(\gamma) \mid \gamma \in R\}$ is an interval of $\mathbf{Z}$, so one can have $\dim(U_{i,s}) = \dim(U_{i+1,s})$ only if $U_{i,s} = e$).

2.1.3. — In the general case, there exists a covering family for the (fpqc) topology, $\{S_j \rightarrow S\}$, such that each pair (G_{S_j}, P_{S_j}) is pinnable (1.14). By the foregoing, one has descent data on the $\text{rad}^u(P_{S_j})_i$, compatible with the vector structures of the quotients, and one concludes by descent of closed subschemes (resp. locally free modules).⁽⁸⁾

Corollary 2.2. *Let S be an affine scheme, G a reductive S -group, and P a parabolic subgroup of G . One has*

$$H^1(S, \text{rad}^u(P)) = 0,$$

i.e. every principal homogeneous bundle under $\text{rad}^u(P)$ is trivial.

Indeed, S decomposes into a sum of subschemes on each of which $\text{rad}^u(P)$ has constant relative dimension. One can therefore, by (iii), suppose that there exists an n such that $U_n = e$. Since $H^1(S, U_i/U_{i+1}) = H^1(S, W(\mathcal{E}_i)) = 0$ by TDTE I, B, 1.1 (or SGA 1, XI 5.1), one concludes immediately.

Corollary 2.3. *Under the preceding conditions, P possesses a Levi subgroup L . If L is a Levi subgroup of P , the canonical map*

$$H^1(S, L) \rightarrow H^1(S, P)$$

is bijective (cf. the introduction to Exp. XXIV for the definition of $H^1(S, \cdot)$).

The first assertion follows from 2.2 and 1.9. The canonical map $H^1(S, L) \rightarrow H^1(S, P)$ is surjective, since P is the semidirect product $L \cdot \text{rad}^u(P)$. To prove that it is injective, it suffices to see that, for every principal homogeneous bundle Q under L , one has $H^1(S, \text{rad}^u(P)_Q) = 0$, where the subscript Q denotes the twisting operation by the L -bundle Q . This can be proved in two ways: one can repeat the proof of 2.2, using the fact that the vector structures on the U_i/U_{i+1} are invariant under L ; one can also note that $\text{rad}^u(P)_Q$ is identified with the unipotent radical of the parabolic subgroup P_Q of G_Q , and apply 2.2 to P_Q .

Corollary 2.4. *Let S be a semilocal scheme, G a reductive S -group, and P a parabolic subgroup of G . There exists a maximal torus T of G contained in P .*

Indeed, by 2.3, P possesses a Levi subgroup L , and it suffices to prove that L possesses a maximal torus, which follows from Exp. XIV, 3.20.

⁽⁸⁾**N.D.E.** Cf. SGA 1, VIII 1.3, 1.9 and 1.10.

Corollary 2.5. *Let S be an affine scheme, G a reductive S -group, and P a parabolic subgroup of G . There exists a locally free $\mathcal{O}_S$ -module $\mathcal{E}$ such that $\mathrm{rad}^u(P)$ is isomorphic as an S -scheme to $W(\mathcal{E})$.*

Indeed, let us prove by induction on i that one has an isomorphism of S -schemes

$$\mathrm{rad}^u(P)/U_i \simeq W(\mathcal{E}_0 \oplus \mathcal{E}_1 \oplus \cdots \oplus \mathcal{E}_{i-1}).$$

This is clear for $i = 0$. Suppose $i > 0$; then $\mathrm{rad}^u(P)/U_i$ is a principal homogeneous bundle with base $\mathrm{rad}^u(P)/U_{i-1}$, under the group

$$(U_{i-1}/U_i)_{\mathrm{rad}^u(P)/U_{i-1}} \simeq W(\mathcal{E}_{i-1} \otimes \mathcal{O}_{\mathrm{rad}^u(P)/U_{i-1}}).$$

Now the base is affine (for example by the induction hypothesis), so this bundle is trivial (TDTE I or SGA 1 XI, *loc. cit.*), and there exists an isomorphism of S -schemes $\mathrm{rad}^u(P)/U_i \simeq (\mathrm{rad}^u(P)/U_{i-1}) \times_S (U_{i-1}/U_i)$, which completes the proof.

Corollary 2.6. *Let S be a semilocal scheme, $\{s_i\}$ the set of its closed points, G a reductive S -group, and P a parabolic subgroup of G . The canonical map*

$$\mathrm{rad}^u(P)(S) \longrightarrow \prod_i \mathrm{rad}^u(P)(\mathrm{Spec} \kappa(s_i))$$

is surjective.

Indeed, if $S = \mathrm{Spec}(A)$, $\kappa(s_i) = A/\mathfrak{p}_i$, and if $\mathcal{E}$ is given by the projective (hence flat) module E , we must prove that the map

$$E \longrightarrow \prod_i E \otimes_A A/\mathfrak{p}_i$$

is surjective. It suffices to do so when $E = A$, in which case this is well known (cf. Bourbaki, *Alg. Comm.* Chap. II, § 1, no. 2, proposition 5).

Corollary 2.7. *Let k be an infinite field, G a reductive k -group, and P a parabolic subgroup of G ; then $\mathrm{rad}^u(P)(k)$ is dense in $\mathrm{rad}^u(P)$.*

Corollary 2.8. *Let S be a semilocal scheme, $\{s_i\}$ the set of its closed points, G a reductive S -group, P a parabolic subgroup of G , and L_i a Levi subgroup of P_{s_i} for each i . There exists a Levi subgroup L of P inducing L_i for each i .*

Indeed, let L_0 be a Levi subgroup of P (2.3). For each i , let $u_i \in \mathrm{rad}^u(P)(\mathrm{Spec}(\kappa(s_i)))$ be such that $\mathrm{int}(u_i)L_{0,s_i} = L_i$ (1.8); if $u \in \mathrm{rad}^u(P)(S)$ induces u_i for each i (2.6), then $L = \mathrm{int}(u)L_0$ has the required property.

Corollary 2.9. *In the situation of 2.1, let moreover H be a group subscheme of G , smooth and of finite presentation over S , with connected fibers, such that $P \cap H$ locally contains a maximal torus of G for the (fpqc) topology. Then, for each $i \geq 0$, there exists a submodule $\mathcal{F}_i$ of $\mathcal{E}_i$, locally a direct summand, such that the isomorphism $U_i/U_{i+1} \xrightarrow{\sim} W(\mathcal{E}_i)$ induces an isomorphism of groups*

$$(U_i \cap H)/(U_{i+1} \cap H) \xrightarrow{\sim} W(\mathcal{F}_i).$$

Indeed, H is a subgroup of type (R) of G (Exp. XXII, 5.2.1). Moreover, the assertion to be proved is local for the (fpqc) topology, and one can suppose G split relative to a maximal torus of $P \cap H$; one can even reduce to the situation of 2.1.1, H being defined by a subset R' of R . Taking up the notations of *loc. cit.*, one sees by Exp. XXII, 5.6.7 (ii) that $U_i \cap H = \prod_{\alpha \in R_i \cap R'} U_\alpha$, hence that $(U_i \cap H)/(U_{i+1} \cap H)$ is identified with $\prod_{\alpha \in R', a(\alpha)=i+1} U_\alpha$, which implies the result.

Corollary 2.10. *In the situation of 2.9, the conclusions of 2.2, 2.5, 2.6, 2.7 are also valid on replacing $\mathrm{rad}^u(P)$ by $\mathrm{rad}^u(P) \cap H$.*

Corollary 2.11. ⁽⁹⁾ *Let G be a reductive S -group, P a parabolic subgroup, and H a subgroup of type (RC) of P such that $\text{rad}^u(H) = \text{rad}^u(P) \cap H$. Then statements 2.2 to 2.8 are also valid on replacing P by H .*

3. Scheme of parabolic subgroups of a reductive group

3.1. — Let E be a finite twisted constant S -scheme (Exp. X, 5.1). Consider the S -functor $\underline{\text{Of}}(E)$, where $\underline{\text{Of}}(E)(S')$ is the set of open and closed subschemes of $E_{S'}$ (or, equivalently, the set of open and closed subsets of $E_{S'}$); then $\underline{\text{Of}}(E)$ is representable by a finite twisted constant S -scheme. Indeed, if $E = A_S$, where A is a finite set, one immediately has $\underline{\text{Of}}(E) \simeq \mathcal{P}(A)_S$ (where $\mathcal{P}(A)$ denotes the set of subsets of A), and one concludes by descent of open and closed subschemes. One evidently has:

$$\underline{\text{Of}}(E_{S'}) = \underline{\text{Of}}(E)_{S'}, \quad \underline{\text{Of}}(E \times_S E') = \underline{\text{Of}}(E) \times_S \underline{\text{Of}}(E').$$

3.2. — Let S be a scheme and G a reductive S -group. The functor $\underline{\text{Par}}(G)$ of parabolic subgroups of G is defined by

$$\underline{\text{Par}}(G)(S') = \text{set of parabolic subgroups of } G_{S'}.$$

In particular $G \in \underline{\text{Par}}(G)(S)$, $\underline{\text{Bor}}(G) \subset \underline{\text{Par}}(G)$. We propose to define a morphism

$$t : \underline{\text{Par}}(G) \longrightarrow \underline{\text{Of}}(\underline{\text{Dyn}}(G))$$

possessing the following properties:

- (i) t is functorial in G (with respect to isomorphisms) and commutes with extension of the base.
- (ii) If $(T, M, R, \Delta, \dots)$ is a pinning of G adapted to the parabolic subgroup P (1.11), the canonical isomorphism $\underline{\text{Dyn}}(G) \simeq \Delta_S$ (Exp. XXIV, 3.4 (iii)) transforms $t(P)$ into $\Delta(P)_S$ (notations of 1.11, 1.12).

First let P be a parabolic subgroup of G , and $(T, M, R, \Delta, \dots)$ a pinning of G adapted to P . One defines $t(P)$ by (ii); the subscheme $t(P)$ of $\underline{\text{Dyn}}(G)$ thus constructed is independent of the pinning chosen. Indeed, if $(T', M', R', \Delta', \dots)$ is another pinning of G adapted to P , the unique inner automorphism of G transforming the first pinning into the second comes from P (1.15); the canonical isomorphism $\Delta \xrightarrow{\sim} \Delta'$ therefore transforms $\Delta(P)$ into $\Delta'(P)$, which implies the stated result.

If one now no longer necessarily supposes (G, P) pinnable, it follows immediately from 1.14 and from the definition of $\underline{\text{Dyn}}(G)$ (Exp. XXIV, 3.3) that one can define by descent a unique open and closed subscheme $t(P)$ of $\underline{\text{Dyn}}(G)$ such that, for every $S' \rightarrow S$ such that $(G, P)_{S'}$ is pinnable, one has $t(P)_{S'} = t(P_{S'})$.

Theorem 3.3. *Let S be a scheme, G a reductive S -group, and*

$$t : \underline{\text{Par}}(G) \longrightarrow \underline{\text{Of}}(\underline{\text{Dyn}}(G))$$

the morphism defined above.

- (i) *In order that two parabolic subgroups P and P' of G be locally conjugate for the (fpqc) topology (cf. 1.3), it is necessary and sufficient that $t(P) = t(P')$.*
- (ii) *$\underline{\text{Par}}(G)$ is representable, and the morphism t is smooth, projective, with integral geometric fibers.*

By 3.2 (i), and the fact that the inner automorphisms of G act trivially on $\underline{\text{Dyn}}(G)$ (Exp. XXIV, 3.4 (iv)), one indeed has $t(P) = t(P')$ when P and P' are conjugate. Conversely, let P and P' be two parabolic subgroups of G such that $t(P) = t(P')$; let us prove that P and P' are conjugate in G , locally for the (fpqc) topology. One can first suppose the pairs (G, P) and (G, P') pinnable (1.14); by conjugacy of pinnings in G (Exp. XXIV, 1.5),

⁽⁹⁾**N.D.E.** Corollary 2.11, which will be used in 4.5.1 and 6.17, has been added.

one can suppose that there exists a pinning (T, M, R, Δ) of G adapted to P and P' . Then $t(P) = t(P')$ implies $\Delta(P) = \Delta(P')$, hence $P = P'$ (cf. 1.4 (v)). We have therefore proved (i). To prove (ii), let us take up the notations of Exp. XXII, 5.11.5.⁽¹⁰⁾

One has a canonical morphism $\underline{\text{Par}}(G) \rightarrow \mathcal{H}_c$, and it is clear (for example by reduction to the pinned case) that it fits into a cartesian square (where the vertical arrows are monomorphisms)

$$\begin{array}{ccc} \underline{\text{Par}}(G) & \xrightarrow{t} & \underline{\text{Of}}(\underline{\text{Dyn}}(G)) \\ \downarrow & & \downarrow \\ \mathcal{H}_c & \xrightarrow{cl} & Cl_c. \end{array}$$

Now (*loc. cit.*) $\mathcal{H}_c$ is representable, and the morphism cl is smooth, quasi-projective, of finite presentation, with integral geometric fibers, so the same holds for t .

It remains to prove that t is proper; but this is now an assertion local for the (fpqc) topology, and one can reduce to the pinned case $G = (G, T, M, R, \Delta, \dots)$. One then has $\underline{\text{Dyn}}(G) \simeq \Delta_S$, and it suffices to prove that, for every subset Δ_1 of Δ , the S -scheme $t^{-1}((\Delta_1)_S)$ is proper over S . Now, if P_1 is the parabolic subgroup of G containing T such that $\Delta(P_1) = \Delta_1$, it follows from (i) that the morphism $G \rightarrow \underline{\text{Par}}(G)$ defined set-theoretically by $g \mapsto \text{int}(g)P_1$ induces an isomorphism from $G/\underline{\text{Norm}}_G(P_1)$ onto $t^{-1}((\Delta_1)_S)$. Now, by 1.2, $G/\underline{\text{Norm}}_G(P_1) = G/P_1$ is projective over S .

Definition 3.4. $\underline{\text{Of}}(\underline{\text{Dyn}}(G))$ is called the *scheme of parabolic types* of G ; $t(P)$ is called the *type* of P .

Corollary 3.5. The S -functor $\underline{\text{Par}}(G)$ is representable by an S -scheme smooth and projective over S . The decomposition

$$\underline{\text{Par}}(G) \rightarrow \underline{\text{Of}}(\underline{\text{Dyn}}(G)) \rightarrow S$$

is the Stein factorization (EGA III, 4.3.3) of the structural morphism $\underline{\text{Par}}(G) \rightarrow S$.

Corollary 3.6. For each $t \in \underline{\text{Of}}(\underline{\text{Dyn}}(G))(S)$, the S -scheme

$$\underline{\text{Par}}_t(G) = t^{-1}(t)$$

of parabolic subgroups of G of type t is smooth and projective over S , homogeneous under G . If P is a parabolic subgroup of G , one has a canonical isomorphism $G/P \xrightarrow{\sim} \underline{\text{Par}}_{t(P)}(G)$. One has $\underline{\text{Par}}_\emptyset(G) = \underline{\text{Bor}}(G)$, $\underline{\text{Par}}_{\underline{\text{Dyn}}(G)}(G) \xrightarrow{\sim} S$.

Remark 3.7. The S -scheme $\underline{\text{Of}}(\underline{\text{Dyn}}(G))$ is equipped with a natural order structure (domination relation, here set-theoretic inclusion, between subschemes). This order structure is a lattice; in particular the infimum of two open and closed subschemes of $\underline{\text{Dyn}}(G)_S$ is evidently their intersection. Note moreover that, if B is a Borel subgroup of G , one can define the functor X of parabolic subgroups of G containing B . The morphism $X \rightarrow \underline{\text{Of}}(\underline{\text{Dyn}}(G))$ induced by t is an isomorphism (for the “ordered scheme” structure), by the assertion $P \subset Q \Rightarrow t(P) \subset t(Q)$ and by:

Lemma 3.8. Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G , and t' a section of $\underline{\text{Of}}(\underline{\text{Dyn}}(G))$ over S such that $t(P) \subset t'$. There exists a unique parabolic subgroup P' of G containing P and such that $t(P') = t'$.

⁽¹⁰⁾**N.D.E.** Recall (cf. *loc. cit.*) that $\mathcal{H}_c = \mathcal{H}_c(G)$ denotes the functor of subgroups of G of type (RC), $Cl_c = Cl_c(G)$ the functor of “conjugacy classes” of such subgroups, and that $cl : \mathcal{H}_c \rightarrow Cl_c$ is the canonical projection.

Extending the base, one can suppose that P contains a Borel subgroup B of G . The uniqueness of P' then follows from 1.17. To prove existence, one can place oneself in the split case, in which case the assertion is evident, cf. no. 1.

Remarks 3.8.1.

- (i) The assertion analogous to 3.8 obtained by reversing the inclusions is evidently false. It would imply, for example, that every group of type A_1 possesses a Borel subgroup, which is not so, cf. Exp. XX, no. 5.
- (ii) It follows immediately from the foregoing that $t(P) \subset t(Q)$ means that, locally for (fpqc) or (ét), P is conjugate to a subgroup of Q (it moreover suffices to verify the assertion on the geometric fibers). Furthermore, we shall see in no. 5 that one can replace the étale topology by the Zariski topology.

3.9. — The preceding discussions can be repeated in the case of critical subgroups. Recall (Exp. XXII, 5.10.4 and 5.10.5) that a reductive subgroup H of the reductive group G is *critical* if $H = \underline{\text{Centr}}_G(\text{rad}(H))$, that a subtorus Q of G is a *C-critical torus* if $Q = \text{rad}(\underline{\text{Centr}}_G(Q))$, and that critical subgroups and C-critical tori⁽¹¹⁾ are in one-to-one correspondence (by $H \mapsto \text{rad}(H)$ and $Q \mapsto \underline{\text{Centr}}_G(Q)$).

If (G, T, M, R) is a split S -group, the subgroup of G containing T corresponding to the subset R' of R (Exp. XXII, 5.4.2) is critical if and only if R' is “vectorial” (that is, the intersection of R with a vector subspace of $M \otimes \mathbf{Q}$), cf. Exp. XXII, 5.10.6.

If G is any reductive S -group, one will define, as in Exp. XXII, 5.11.5, a finite étale S -scheme $C\ell_{\text{crit}}$, which in the split case will be the constant scheme associated with the set of vectorial subsets of R modulo the action of the Weyl group. If $\underline{\text{Crit}}(G)$ denotes the “functor of critical subgroups” of G , one will have a canonical morphism $\underline{\text{Crit}}(G) \xrightarrow{cl} C\ell_{\text{crit}}$, which will fit into a cartesian diagram⁽¹²⁾

$$\begin{array}{ccc} \underline{\text{Crit}}(G) & \xrightarrow{cl} & C\ell_{\text{crit}} \\ \downarrow & & \downarrow \\ \mathcal{H}_c & \xrightarrow{cl} & C\ell_c. \end{array}$$

Proposition 3.10. *Let S be a scheme, G a reductive S -group, $\underline{\text{Crit}}(G)$ the functor of its critical subgroups, and $C\ell_{\text{crit}}$ and $cl : \underline{\text{Crit}}(G) \rightarrow C\ell_{\text{crit}}$ the finite étale S -scheme and the morphism defined above.*

- (i) *In order that the critical subgroups H and H' of G be conjugate (locally for the (fpqc) topology), it is necessary and sufficient that $cl(H) = cl(H')$.*
- (ii) *$\underline{\text{Crit}}(G)$ is representable, and the morphism cl is smooth, affine, with integral geometric fibers.*

This is proved as in 3.3, except for the assertion “ cl is affine”. It suffices to prove that $\underline{\text{Crit}}(G)$ is affine over S . Now $\underline{\text{Crit}}(G)$ is naturally identified with the S -functor of critical tori of G , and one therefore has a canonical monomorphism

$$\underline{\text{Crit}}(G) \rightarrow M,$$

where M is the scheme of subgroups of multiplicative type of G (Exp. XI, 4.1). To prove that $\underline{\text{Crit}}(G)$ is affine over S , it suffices, by Exp. XII 5.3, to show that this morphism is an open and closed immersion, or again, making the base change $M \rightarrow S$, to prove the following assertion: if Q is a subgroup of multiplicative type of the reductive group G , the $S' \rightarrow S$

⁽¹¹⁾**N.D.E.** Here “critical torus” has been replaced by “C-critical torus”, cf. *loc. cit.*. In the sequel, one will simply write “critical torus” instead of “C-critical torus”.

⁽¹²⁾**N.D.E.** Cf. 3.3, N.D.E. (10), for the notations $\mathcal{H}_c$ and $C\ell_c$.

such that $Q_{S'}$ is a critical torus of $G_{S'}$ are those which factor through a certain open and closed subscheme of S .

Now saying that Q is a critical torus means:

- (1) that Q is a torus,
- (2) Q being a torus, that $\text{rad}(\text{Centr}_G(Q))$, which is also a torus, has the same relative dimension as Q .

Now these two conditions are indeed of the type considered.

Corollary 3.11. *The S -functor $\text{Crit}(G)$ is representable by an S -scheme smooth and affine over S .*

Corollary 3.12. *Let H be a critical subgroup of the reductive S -group G . Then $G/\text{Norm}_G(H)$ and G/H are representable by S -schemes affine and smooth over S .*

The first assertion follows from 3.11, the second from the first and from Exp. XXII, 5.10.2.

Corollary 3.13. *Let Q be a subtorus of the reductive S -group G . Then $G/\text{Centr}_G(Q)$ is representable by an S -scheme smooth and affine over S . The same holds for $G/\text{Norm}_G(Q)$ if Q is a critical subtorus of G .*

Indeed, $H = \text{Centr}_G(Q)$ is critical (Exp. XXII, 5.10.5), and one has $\text{Norm}_G(H) = \text{Norm}_G(Q)$ if Q is critical (*loc. cit.*, 5.10.8).

3.14. — By conjugacy of the Levi subgroups of the parabolic subgroups of G , there exists a unique morphism

$$u : \text{Of}(\text{Dyn}(G)) \longrightarrow C\ell_{\text{crit}}$$

such that, for every parabolic subgroup P of G and every Levi subgroup L of P , one has $c\ell(L) = u(t(P))$, and such that this holds after every base change.

3.15. — Let S be a scheme and G a reductive S -group. Consider the S -functors:⁽¹³⁾

$$\begin{aligned} \text{PL}(S') &= \{\text{pairs } P \supset L, P \text{ parabolic in } G_{S'}, \\ &\quad L \text{ a Levi subgroup of } P\}; \\ \text{PT}(S') &= \{\text{pairs } P \supset T, P \text{ parabolic in } G_{S'}, \\ &\quad T \text{ a maximal torus of } P\}; \\ \text{CT}(S') &= \{\text{pairs } C \supset T, C \text{ a critical subgroup of } G_{S'}, \\ &\quad T \text{ a maximal torus of } C\}; \\ \text{PLT}(S') &= \{\text{triples } P \supset L \supset T, (P, T) \in \text{PT}(S'), \\ &\quad L \text{ a Levi subgroup of } P\}. \end{aligned}$$

One has evident morphisms between these functors and the functors $\text{Par}(G)$, $\text{Crit}(G)$, $\text{Tor}(G)$ already introduced, and one has a commutative diagram in the shape of a truncated cube (see the following figure):

⁽¹³⁾**N.D.E.** LT has been changed to CT .

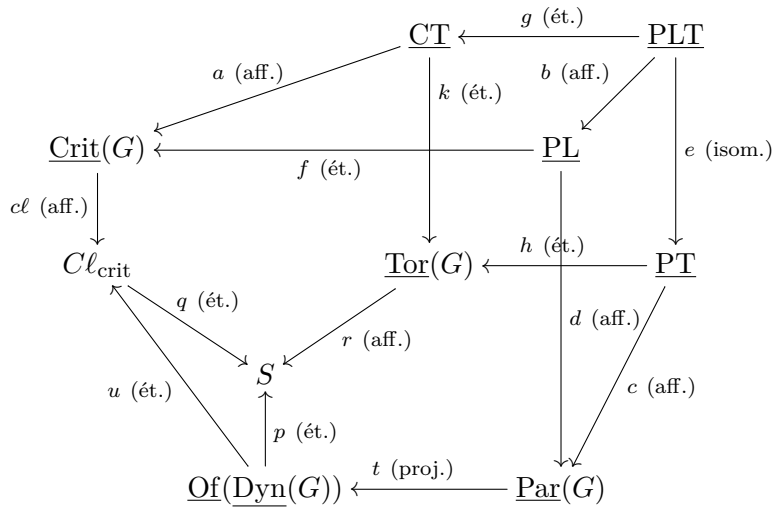


Figure 3.15.1

Theorem 3.16. (*Cf. figure 3.15.1.*)

- (i) All the morphisms of the diagram are smooth, surjective, and of finite presentation.
- (ii) All the morphisms of the diagram, except t , are affine; the morphism t is projective.
- (iii) All the morphisms of the diagram are either finite étale or have integral geometric fibers: the morphisms f, g, h, k, p, q and u are finite étale, the morphisms a, b, c, d, r and $c\ell$ have integral geometric fibers, and the morphism e is an isomorphism.
- (iv) The square (a, b, f, g) is cartesian.

PROOF. First it is clear that e is an isomorphism, by 1.6 (ii). Moreover, (iv) is evident.

The morphism a is smooth, affine, with integral geometric fibers: indeed, by the base change $\text{Crit}(G) \rightarrow S$, it suffices to verify that the morphism $\text{Tor}(L_0) \rightarrow S$, where L_0 is the universal critical subgroup, possesses these properties; now L_0 is reductive (by definition), and one is reduced to Exp. XXII, 5.8.3. The morphism b therefore possesses the same properties, by (iv).

The morphism d is also smooth, affine, with integral geometric fibers, by 1.9; the same therefore holds for $c = dbe^{-1}$. The morphism r possesses these same properties (Exp. XXII, 5.8.3), as does the morphism $c\ell$ (3.10).

Moreover, we have already proved that the morphisms p and q are finite étale surjective (3.1 and 3.9). If we prove that f and k are finite étale surjective, the same properties will hold for g (by (iv)) and for h (since $h = kge^{-1}$); since the stated properties of t were proved in 3.3 (ii), it therefore remains only to prove that f (resp. k) is finite étale surjective; let us give the proof for k , the one for f being analogous.

It suffices for us to prove that, if T is a maximal torus of G , the functor $\underline{C}$ of critical subgroups of G containing T is representable by a finite étale S -scheme with nonempty fibers; one can suppose G split relative to T ; let then E be the set of vectorial subsets of R (the root system of the splitting); $\underline{C}$ is representable by E_S (3.9), which completes the proof. $\square$

Corollary 3.17. *All the functors of the preceding diagram are representable by S -schemes smooth over S , and they are all affine over S , except for $\text{Par}(G)$.*

Remark 3.18.

- (i) The fact that the morphism $f : \text{PL} \rightarrow \text{Crit}(G)$ is étale surjective implies that a subgroup of G is critical if and only if it is, locally for the étale topology, a Levi subgroup of a parabolic subgroup of G .

On the other hand, one should not believe that in general the map $f(S) : \text{PL}(S) \rightarrow \text{Crit}(G)(S)$ is surjective: it may very well happen that a critical subgroup

C of G does not come over S from a parabolic subgroup of G ; for example, a maximal torus is not always contained in a Borel subgroup (example: a nonsplit form of SL_2 , cf. Exp. XX, no. 5).

- (ii) Similarly, it may happen that the morphism $u : \underline{\mathrm{Of}}(\underline{\mathrm{Dyn}}(G)) \longrightarrow C\ell_{\mathrm{crit}}$ is not an isomorphism: two parabolic subgroups of distinct types may have Levi subgroups of the same type; example: in a group of type A_2 , there are two types of parabolic subgroups whose Levi subgroups have semisimple rank 1 (corresponding to the two vertices of the diagram), whereas there is only one type of critical subgroups of rank 1.

The analogous example with a group of type A_3 shows that, even over an algebraically closed field, nonisomorphic parabolic subgroups may have Levi subgroups of the same type.⁽¹⁴⁾

Let us end this section with an application to the theory of principal bundles.

Lemma 3.20. ⁽¹⁵⁾ *Let S be a scheme, G a reductive S -group, F a principal homogeneous bundle under G , and G^F the corresponding twisted form of G . Identify $\underline{\mathrm{Dyn}}(G)$ and $\underline{\mathrm{Dyn}}(G^F)$ (Exp. XXIV, 3.5). Let P be a parabolic subgroup of G . One has a canonical isomorphism*

$$F/P \xrightarrow{\sim} \underline{\mathrm{Par}}_t(P)(G^F).$$

In particular, in order that the structural group of F can be reduced to P , it is necessary and sufficient that G^F possess a parabolic subgroup of type $t(P)$.

The proof proceeds exactly as in Exp. XXIV, 4.2.1.

3.21. — If S is a scheme, G a reductive S -group, and if $t \in \underline{\mathrm{Of}}(\underline{\mathrm{Dyn}}(G))(S)$, one denotes by $H_t^1(S, G)$ the subset of $H^1(S, G)$ consisting of the classes of principal bundles F under G such that the associated group G^F possesses a parabolic subgroup of type t . If G itself possesses a parabolic subgroup P of type t , $H_t^1(S, G)$ is precisely the image of $H^1(S, P)$ in $H^1(S, G)$, an image which therefore does not depend on the chosen P .

4. Relative position of two parabolic subgroups

4.1. A preliminary result

Lemma 4.1.1. *Let k be a field, G a reductive k -group, and P and P' two parabolic subgroups of G .*

- (i) *Then $P \cap P'$ is smooth, has the same reductive rank as G , and contains a maximal torus T of G .*

⁽¹⁴⁾**N.D.E.** I.e. the parabolic subgroups

$$P = \left\{ \begin{pmatrix} * & * & * & * \\ * & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{pmatrix} \right\}, \quad P' = \left\{ \begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & 0 & 0 & * \end{pmatrix} \right\}$$

of GL_4 are not isomorphic (since $\mathrm{rad}^u(P) \not\cong \mathrm{rad}^u(P')$), but their Levi subgroups

$$L = \left\{ \begin{pmatrix} * & * & 0 & 0 \\ * & * & 0 & 0 \\ 0 & 0 & * & 0 \\ 0 & 0 & 0 & * \end{pmatrix} \right\}, \quad L' = \left\{ \begin{pmatrix} * & 0 & 0 & 0 \\ 0 & * & * & 0 \\ 0 & * & * & 0 \\ 0 & 0 & 0 & * \end{pmatrix} \right\}$$

are conjugate by the element $\begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$.

⁽¹⁵⁾**N.D.E.** The numbering of the original has been retained: there is no no. 3.19.

- (ii) *The set R' of roots of $P \cap P'$ relative to T is a closed subset of the set R of roots of G relative to T .⁽¹⁶⁾*

First suppose k algebraically closed. Let B (resp. B') be a Borel subgroup of P (resp. P'). One knows that there exists $g \in G(k)$ such that $\text{int}(g)B = B'$. Moreover, if T_0 is a maximal torus of B , and if one puts $N = \underline{\text{Norm}}_G(T_0)$, one knows (Bruhat's theorem, Bible, § 13.4, cor. 1 to th. 3) that $G(k) = B(k)N(k)B(k)$. One therefore sees that there exist $b, b_1 \in B(k)$ and $n \in N(k)$ such that $g = bnb_1$, hence $\text{int}(b)\text{int}(n)B = B'$. One then has

$$P \cap P' \supset B \cap B' = \text{int}(b)(B \cap \text{int}(n)B') \supset \text{int}(b)T_0.$$

Now suppose k arbitrary. Applying the preceding result, one sees that $P_{\bar{k}} \cap P'_{\bar{k}}$ contains a maximal torus of $G_{\bar{k}}$; by Exp. XXII, 5.4.5, one deduces that $(P \cap P')_{\bar{k}}$ is smooth “along the identity section”, hence smooth since one is over a field (Exp. VIA 1.3.1), hence that $P \cap P'$ is smooth. By Exp. VIA 2.3.1, the identity component $(P \cap P')^0$ of $P \cap P'$ is therefore an open subgroup of $P \cap P'$, smooth over S . One can then apply Exp. XIV, 1.1 to it, whence (i).

Finally, the set R_P (resp. $R_{P'}$) of roots of P (resp. P') relative to T is closed, and one has $R' = R_P \cap R_{P'}$, whence (ii).

Remark 4.1.2. One can prove ([BT65], 4.5) that $P \cap P'$ is connected;⁽¹⁷⁾ this will be used in 4.5.1.

Remark 4.1.3. The preceding lemma does not hold over an arbitrary scheme. Indeed, let for example G be a reductive group over an algebraically closed field k , and let B be a Borel group of G . Take $G = X$ as base, and consider the Borel subgroups B_1 and B_2 of G_X , where $B_1 = B_X$ and $B_2 = \text{int}(g_0)B_1$, g_0 being the canonical (diagonal) section of G_X . For each $g \in X(k)$, the fiber of $B_1 \cap B_2$ at g is precisely $B \cap \text{int}(g)B$. If one supposes $B \neq G$, the dimension of this fiber varies with g , so $B_1 \cap B_2$ cannot be smooth over X .

4.2. Transversal position

Theorem 4.2.1. *Let S be a scheme, G a reductive S -group, and P and Q two parabolic subgroups of G . The following conditions on the pair (P, Q) are equivalent:*

- (i) $\text{Lie}(P/S) + \text{Lie}(Q/S) = \text{Lie}(G/S)$.
- (ii) *The canonical morphism $P \times_S Q \rightarrow G$ is smooth.*
- (ii') *The canonical morphism $P \rightarrow G/Q$ is smooth.*
- (iii) *The canonical morphism $P \times_S Q \rightarrow G$ is open.*
- (iii') *The canonical morphism $P \rightarrow G/Q$ is open.*
- (iv) *The canonical morphism $P \times_S Q \rightarrow G$ is dominant fiber by fiber.*
- (iv') *The canonical morphism $P \rightarrow G/Q$ is dominant fiber by fiber.*
- (v) *For every $s \in S$, “ $P_{\bar{s}} \cap Q_{\bar{s}}$ has minimum dimension”, i.e. one has*

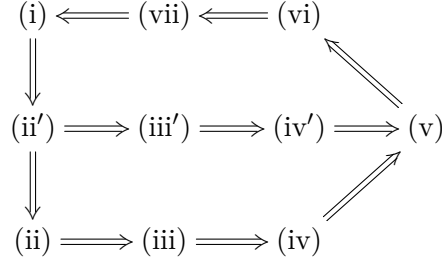
$$\dim(P_{\bar{s}} \cap Q_{\bar{s}}) = \dim P_{\bar{s}} + \dim Q_{\bar{s}} - \dim G_{\bar{s}}.$$

- (vi) *There exist a covering family for the étale topology $\{S_i \rightarrow S\}$, and for each i a Borel subgroup B_i of P_{S_i} and a Borel subgroup B'_i of Q_{S_i} such that $B_i \cap B'_i$ is a maximal torus of G_{S_i} .*
- (vii) *There exist a covering family for the étale topology $\{S_i \rightarrow S\}$, and for each i a splitting $(T_i, \dots, R_i)$ of G_{S_i} and a system of positive roots R_i^+ of R_i such that P_{S_i} (resp. Q_{S_i}) is the subgroup of type (R) of G_{S_i} containing T_i and defined by a subset $R_i^{(1)}$ (resp. $R_i^{(2)}$) of R containing R_i^+ (resp. $-R_i^+$) (see Exp. XXII, 5.4.2 and 5.2.1 for the definitions).*

⁽¹⁶⁾**N.D.E.** Part (ii), which will be useful in 4.5.1, has been added.

⁽¹⁷⁾**N.D.E.** In view of the details added in 4.5.1, the original (which said “we shall not use this fact”) has been modified here.

PROOF. We shall prove the theorem according to the logical diagram



One trivially has $(ii) \Rightarrow (iii)$ and $(ii') \Rightarrow (iii')$. If (iii) holds, the set-theoretic image of the morphism $P \times_S Q \Rightarrow G$ is an open subset of G which contains the identity section; since the fibers of G are connected, this image is dense on each fiber, which proves (iv) . Similarly, one has $(iii') \Rightarrow (iv')$.

One has $(ii') \Rightarrow (ii)$, by the cartesian diagram

$$\begin{array}{ccc}
 P \times_S Q & \longrightarrow & G \\
 \text{pr}_1 \downarrow & & \downarrow \\
 P & \longrightarrow & G/Q.
 \end{array}$$

Moreover, (iv) or (iv') implies (v) , by dimension theory (cf. EGA IV₂, 5.6.6). One notes that one can indeed suppose $S = \text{Spec}(k)$, k an algebraically closed field, and that every nonempty fiber of the morphism (iv) , resp. (iv') , at a point of $G(k)$, resp. of $(G/Q)(k)$, is isomorphic to $P \cap Q$ (as an immediate computation shows).

One has $(vi) \Rightarrow (vii)$, by Exp. XXII, 5.5.1 (iv) and 5.9.2.

One has $(vii) \Rightarrow (i)$, since, to verify that $\text{Lie}(P) + \text{Lie}(Q) = \text{Lie}(G)$, one can reason locally for (fpqc); hence, if (vii) holds, one can suppose G split, $P \supset B_{R_+}$ and $Q \supset B_{-R_+}$ (usual notations), in which case one already has

$$\text{Lie}(B_{R_+}) + \text{Lie}(B_{-R_+}) = \text{Lie}(G).$$

Let us prove that (i) implies (ii') .

Let $u : P \rightarrow G/Q$ be the canonical morphism; to prove that u is smooth, one can content oneself with doing so for the geometric fibers of u , since P and G/Q are smooth over S , and one can therefore suppose that S is the spectrum of an algebraically closed field. Since the morphism u is compatible with the evident action of P (one has $u(pp') = pu(p')$), it suffices, by a translation argument (cf. the proof of VIB 1.3), to verify that u is smooth at $e \in P(k)$, i.e. (SGA 1, II 4.7) that the tangent map to u at e is surjective; but this is naturally identified with the canonical map $\text{Lie}(P) \rightarrow \text{Lie}(G)/\text{Lie}(Q)$, which is surjective if (i) holds.

It therefore remains only to verify the last assertion, that is, $(v) \Rightarrow (vi)$. First suppose that S is the spectrum of an algebraically closed field. By 4.1.1, there exists a maximal torus T contained in P and Q ; let R (resp. R_1 , resp. R_2) be the set of roots of G (resp. P , resp. Q) relative to T . One has:

$$\begin{aligned}
 \dim(G) &= \dim(T) + \text{Card}(R), & \dim(P) &= \dim(T) + \text{Card}(R_1), \\
 \dim(Q) &= \dim(T) + \text{Card}(R_2), & \dim(P \cap Q) &= \dim(T) + \text{Card}(R_1 \cap R_2),
 \end{aligned}$$

by Exp. XXII, 5.4.4 and 5.4.5, for example. Condition (v) is therefore equivalent to

$$\text{Card}(R_1 \cap R_2) = \text{Card}(R_1) + \text{Card}(R_2) - \text{Card}(R),$$

that is, $R_1 \cup R_2 = R$. To prove (vi) , it suffices, by Exp. XXII, 5.9.2 and 5.4.5, to prove that $R_1 \cap -R_2$ contains a system of positive roots of R . One is therefore reduced to proving:

Lemma 4.2.2. *Let R be a “root system” (for example the set of roots of a root datum in the sense of Exposé XXI). Let R_1 and R_2 be two closed subsets of R , each containing a system of positive roots. If $R_1 \cup R_2 = R$, then $R_1 \cap -R_2$ contains a system of positive roots.*

Indeed, since $R_1 \cap -R_2 = R_3$ is evidently closed, and by Exp. XXI, 3.3.6, it suffices to show that $R_3 \cup -R_3 = R$. Now one knows that $R_1 \cup -R_1 = R = R_2 \cup -R_2$, and one concludes by the following elementary fact: if A, A', B, B' are four subsets of a set E , and if $A \cup A' = B \cup B' = A \cup B = A' \cup B' = E$, one has $(A \cap B') \cup (A' \cap B) = E$.

This completes the proof of (v) $\Rightarrow$ (vi) in the case where the base is the spectrum of an algebraically closed field. Let us now return to the general case and suppose (v) satisfied. Let $s \in S$; by the foregoing, one can find a Borel subgroup $\overline{B}$ (resp. $\overline{B}'$) of $P_{\overline{s}}$ (resp. $Q_{\overline{s}}$) such that $\overline{B} \cap \overline{B}'$ is a maximal torus of $G_{\overline{s}}$.

Since the S -scheme $\underline{\text{Bor}}(P) \simeq \underline{\text{Bor}}(P/\text{rad}^u(P))$ of Borel subgroups of P is smooth, one can, applying “Hensel’s lemma” (cf. Exp. XI 1.10) and reasoning locally for the étale topology (i.e. replacing S by an étale $S' \rightarrow S$ covering s , and s by a point of its fiber in S'), suppose that there exists a Borel subgroup B of P projecting onto $\overline{B}$; one can similarly suppose that there exists a Borel subgroup B' of Q projecting onto $\overline{B}'$. Since $\overline{B} \cap \overline{B}'$ is a maximal torus of G , there exists an open subset U of S containing s and such that $B_U \cap B'_U$ is a maximal torus of G_U (Exp. XXII, 5.9.4), which proves (vi). $\square$

Definition 4.2.3. A pair (P, Q) satisfying the equivalent conditions (i) to (vii) of theorem 4.2.1 is said to be *in transversal position*. One also says that P is *in transversal position relative to Q* , or, by abuse of language, that P and Q are in (mutual) transversal position.

By (vi), this definition coincides, in the case of Borel subgroups, with that of Exp. XXII, 5.9.1.

Corollary 4.2.4. *Let S be a scheme, G a reductive S -group, and P and Q two parabolic subgroups of G .*

- (i) *In order that (P, Q) be in transversal position, it is necessary and sufficient that, for each point s of S , the pair $(P_{\overline{s}}, Q_{\overline{s}})$ be in transversal position; if $S' \rightarrow S$ is a surjective morphism, and if $(P_{S'}, Q_{S'})$ is in transversal position, then (P, Q) is in transversal position.*
- (ii) *There exists an open subscheme U of S satisfying the following property: in order that a morphism $S' \rightarrow S$ factor through U , it is necessary and sufficient that $(P_{S'}, Q_{S'})$ be in transversal position.*
- (iii) *Consider the subfunctors*

$$\begin{aligned} \underline{\text{Gen}}(G) &\subset \underline{\text{Par}}(G) \times_S \underline{\text{Par}}(G), \\ \underline{\text{Gen}}(/Q) &\subset \underline{\text{Par}}(G), \\ \underline{\text{Gen}}(P/Q) &\subset G, \end{aligned}$$

defined as follows: for $S' \rightarrow S$, $\underline{\text{Gen}}(G)(S')$ is the set of pairs of parabolic subgroups of $G_{S'}$ in transversal position, $\underline{\text{Gen}}(/Q)(S')$ is the set of parabolic subgroups of $G_{S'}$ in transversal position relative to $Q_{S'}$, and $\underline{\text{Gen}}(P/Q)(S')$ is the set of $g \in G(S')$ such that $\text{int}(g)P_{S'}$ is in transversal position relative to $Q_{S'}$.

Each of these functors is representable by an open subscheme universally schematically dense over $S^{(18)}$ (cf. Exp. XVIII § 1) of the corresponding S -scheme $\underline{\text{Par}}(G) \times_S \underline{\text{Par}}(G)$, resp. $\underline{\text{Par}}(G)$, resp. G .

Assertions (i) follow immediately from the description 4.2.1 (i) of the term “transversal position”. To prove (ii), one takes $U = S - \text{Supp}(\text{Coker } u)$, where u is the canonical morphism $\text{Lie}(P) \oplus \text{Lie}(G) \rightarrow \text{Lie}(G)$.

⁽¹⁸⁾**N.D.E.** “Relatively dense” has been replaced by “universally schematically dense over S ”, cf. EGA IV₃, Def. 11.10.8.

Since one has cartesian diagrams

$$\begin{array}{ccc} G & \xrightarrow{f} & \underline{\text{Par}}(G) \\ \uparrow & & \uparrow \\ \underline{\text{Gen}}(P/Q) & \longrightarrow & \underline{\text{Gen}}(/Q) \end{array} \quad \begin{array}{ccc} \underline{\text{Par}}(G) & \xrightarrow{f'} & \underline{\text{Par}}(G) \times_S \underline{\text{Par}}(G) \\ \uparrow & & \uparrow \\ \underline{\text{Gen}}(/Q) & \longrightarrow & \underline{\text{Gen}}(G) \end{array}$$

(where $f(g) = \text{int}(g)P$ and $f'(R) = (R, Q)$), it suffices to verify (iii) in the case of $\underline{\text{Gen}}(G)$. Let then P_0 be the canonical parabolic subgroup of $G_{\underline{\text{Par}}(G)}$; put

$$X = \underline{\text{Par}}(G) \times_S \underline{\text{Par}}(G), \\ P = \text{pr}_1^*(P_0), \quad Q = \text{pr}_2^*(P_0);$$

applying assertion (ii) to the parabolic subgroups P and Q of G_X , one constructs an open subscheme U of X which, as one immediately verifies, is indeed identified with $\underline{\text{Gen}}(G)$. It remains to verify the density assertion, which can be done on the geometric fibers;⁽¹⁹⁾ one can therefore suppose $S = \text{Spec}(k)$, k an algebraically closed field; since $\underline{\text{Par}}(G)$ is smooth, it suffices to verify that $\underline{\text{Gen}}(G)$ meets every irreducible component of $\underline{\text{Par}}(G) \times_S \underline{\text{Par}}(G)$; in other words, by 3.3, it suffices to see that, if $t, t' \in \text{Of}(\underline{\text{Dyn}}(G))(S)$, there exists a pair (P, P') in transversal position, with $t(P) = t$, $t(P') = t'$.

Now this is immediate: one chooses a pair (B, B') of Borel subgroups of G such that $B \cap B'$ is a maximal torus (one splits G and applies Exp. XXII, 5.9.2), then applies 3.8 to construct $P \supset B$ and $P' \supset B'$, with $t(P) = t$, $t(P') = t'$; P and P' have the desired types and are in transversal position by 4.2.1 (vi).

Corollary 4.2.5. *Let S be a scheme, G a reductive S -group, and P and Q two parabolic subgroups of G , the pair (P, Q) being in transversal position.*

- (i) *Let P' and Q' be two parabolic subgroups of G of the same types as P and Q respectively. In order that the pair (P', Q') be in transversal position, it is necessary and sufficient that it be conjugate to the pair (P, Q) , locally for the étale topology. (N.B. We shall see in § 5 that one can replace the étale topology by the Zariski topology.)*
- (ii) *The canonical morphism $P \times_S Q \longrightarrow G$ induces a smooth surjective morphism $P \times_S Q \longrightarrow \underline{\text{Gen}}(Q/P)$, and an isomorphism*

$$(P \times_S Q)/(P \cap Q) \simeq \underline{\text{Gen}}(Q/P)$$

(where $P \cap Q = R$ acts on $P \times_S Q$ by $(p, q)r = (pr, r^{-1}q)$).

- (iii) *The canonical morphism $P \longrightarrow \underline{\text{Par}}_{t(Q)}(G)$ (defined set-theoretically by $p \longmapsto \text{int}(p)Q$) induces a smooth surjective morphism $P \longrightarrow \underline{\text{Gen}}(/P) \cap \underline{\text{Par}}_{t(Q)}(G)$, and an isomorphism*

$$P/(P \cap Q) \simeq \underline{\text{Gen}}(/P) \cap \underline{\text{Par}}_{t(Q)}(G).$$

- (iv) *The canonical morphism $G \longrightarrow \underline{\text{Par}}_{t(P)}(G) \times_S \underline{\text{Par}}_{t(Q)}(G)$ (defined set-theoretically by $g \longmapsto (\text{int}(g)P, \text{int}(g)Q)$) induces a smooth surjective morphism*

$$G \longrightarrow \underline{\text{Gen}}(G) \cap (\underline{\text{Par}}_{t(P)}(G) \times_S \underline{\text{Par}}_{t(Q)}(G))$$

and an isomorphism

$$G/(P \cap Q) \simeq \underline{\text{Gen}}(G) \cap (\underline{\text{Par}}_{t(P)}(G) \times_S \underline{\text{Par}}_{t(Q)}(G)).$$

Let us prove (i). It is clear that the condition is sufficient; let us prove that it is necessary. Let therefore (P', Q') be in transversal position. Since P and P' are locally conjugate for the étale topology, one can suppose $P = P'$, and it suffices for us to prove that, if Q and Q' are two parabolic subgroups of G in transversal position relative to P and of the same type, they

⁽¹⁹⁾**N.D.E.** Cf. EGA IV₃, 11.10.10.

are then conjugate, locally for the étale topology, by a section of P . Using 4.2.1 (vi), one can suppose that there exist Borel subgroups B, B', B_1, B'_1 of P, P, Q, Q' respectively such that $B \cap B_1 = T$ and $B' \cap B'_1 = T'$ are maximal tori of G . Now the Killing pairs (B, T) and (B', T') of P are locally conjugate in P for the étale topology (1.16), and one can suppose $B = B', T = T'$, in which case one has $B_1 = B'_1$ by Exp. XXII, 5.9.2, hence $Q = Q'$ by 3.8.

Assertions (ii), (iii) and (iv) are proved in parallel fashion. Let us prove (ii), for example; let $g \in \text{Gen}(Q/P)(S)$, i.e. let $g \in G(S)$ be such that $\text{int}(g)Q$ is in transversal position relative to P . By the preceding proof, Q and $\text{int}(g)Q$ are locally conjugate for the étale topology by a section of P .

Reasoning locally for this topology, one can suppose that there exists $p \in P(S)$ such that $\text{int}(g)Q = \text{int}(p)Q$, hence $p^{-1}g \in \text{Norm}_G(Q)(S) = Q(S)$; this proves the existence of a $q \in Q(S)$ such that $g = pq$. We have therefore proved that the morphism considered in (ii) is covering for the étale topology. Comparing with 4.2.1 (ii), one deduces that it is smooth and surjective. Moreover, an immediate argument shows that the equivalence relation defined on $P \times_S Q$ by the morphism $P \times_S Q \rightarrow G$ is the equivalence relation associated with the action of the group $R = P \cap Q$ (acting by $(p, q)r = (pr, r^{-1}q)$), which proves the last assertion of (ii) (since a smooth surjective morphism is an effective epimorphism, for example).

Remark 4.2.6. If P and Q are in transversal position, one will often denote by $P \cdot Q$ the open subset $\text{Gen}(Q/P)$ of G , a notation justified by 4.2.5 (ii).

Proposition 4.2.7. *Let S be a scheme, G a reductive S -group, and P and Q two parabolic subgroups of G , the pair (P, Q) being in transversal position.*

- (i) *The group $P \cap Q$ is smooth over S (and in fact has connected fibers by 4.1.2); let us then introduce $(P \cap Q)^0$ (cf. Exp. VIB 3.10); this is a subgroup of type (RC) of G (Exp. XXII, 5.11.1), whose unipotent radical (loc. cit., 5.11.4) decomposes as a direct product*

$$\text{rad}^u((P \cap Q)^0) = (\text{rad}^u(P) \cap Q) \times_S (P \cap \text{rad}^u(Q)).$$

- (ii) *If S is affine, $H^1(S, \text{rad}^u(P \cap Q)^0) = 0$. If S is semilocal, $P \cap Q$ contains a maximal torus of G .*

Indeed, $P \cap Q$ is smooth by 4.2.1 (ii) and the cartesian diagram

$$\begin{array}{ccc} P \times_S Q & \longrightarrow & G \\ \uparrow & & \uparrow \\ P \cap Q & \longrightarrow & S. \end{array}$$

To verify the stated assertions about $(P \cap Q)^0$, one can reason locally for the étale topology, hence by 4.2.1 (vii) suppose that one has chosen a splitting (G, T, M, R) of G such that P and Q contain T and are defined respectively by subsets R_1 and R_2 of R , R_1 containing a system of positive roots R_+ , and R_2 containing the opposite system R_- . Let Δ be the set of simple roots of R_+ ; put

$$A_1 = \Delta \cap -R_1, \quad A_2 = \Delta \cap R_2, \quad A = A_1 \cap A_2.$$

By 1.4 (v) and 1.12, one has

$$\begin{aligned} R_1 &= R_+ \cup (R_- \cap -\mathbf{N}A_1), & R_2 &= (R_+ \cap \mathbf{N}A_2) \cup R_-, \\ \text{rad}^u(P) &= \prod_{\substack{\alpha \in R_1 \\ \alpha \notin -R_1}} U_\alpha, & \text{rad}^u(Q) &= \prod_{\substack{\alpha \in R_2 \\ \alpha \notin R_2}} U_\alpha. \end{aligned}$$

By Exp. XXII, 5.6.7, one therefore has

$$\begin{aligned}\mathrm{rad}^u(P) \cap Q &= \prod_{\substack{\alpha \in R_1 \cap R_2 \\ \alpha \notin -R_1}} U_\alpha = \prod_{\alpha \in K_2} U_\alpha, \\ \mathrm{rad}^u(Q) \cap P &= \prod_{\substack{\alpha \in R_1 \cap R_2 \\ \alpha \notin -R_2}} U_\alpha = \prod_{\alpha \in K_1} U_\alpha,\end{aligned}$$

where K_2 is the set of positive roots which are linear combinations of the elements of A_2 , but not linear combinations of the elements of A , and K_1 the set of negative roots which are linear combinations of the elements of A_1 , but not linear combinations of the elements of A . It is clear that, if $\alpha \in K_2$, $\beta \in K_1$, $\alpha + \beta$ is never a root, nor zero, which implies that the two groups above commute.

Moreover, one knows by Exp. XXII, 5.4.5 that $H = (P \cap Q)^0$ is defined by the set of roots $R_1 \cap R_2$, namely

$$R_1 \cap R_2 = (R_+ \cap \mathbf{N}A_2) \cup (R_- \cap -\mathbf{N}A_1).$$

Since $R_1 \cap R_2$ is closed, H is of type (RC) by definition (Exp. XXII 5.11.1), and by *loc. cit.*, 5.11.3 and 5.11.4, one has

$$\mathrm{rad}^u(H) = \prod_{\alpha \in K} U_\alpha,$$

where K is the set of $\alpha \in R_1 \cap R_2$ such that $\alpha \notin -(R_1 \cap R_2)$. Since the symmetric part of $R_1 \cap R_2$ is evidently $R \cap \mathbf{Z}A$, one sees immediately that $K = K_1 \cup K_2$, which ends the proof of (i).

The first assertion of (ii) then follows from (i) and 2.10; let us prove the second. Since $(P \cap Q)^0 / \mathrm{rad}^u((P \cap Q)^0)$ is reductive, it possesses a maximal torus $\overline{T}$ if the base is semilocal. The inverse image of $\overline{T}$ in $(P \cap Q)^0$ is a subgroup N of type (R) of G with solvable fibers, and one has $N^u = \mathrm{rad}^u((P \cap Q)^0)$ (Exp. XXII, 5.6.9). The scheme of maximal tori of N is a principal homogeneous bundle under N^u (*loc. cit.*, 5.6.13), and so possesses a section, since $H^1(S, N^u) = 0$.

4.3. Opposite parabolic subgroups

4.3.1. — If G is a reductive S -group, one defined in Exp. XXIV, 3.16.6 a canonical “outer automorphism” s_G of order $\leq 2^{(20)}$ of G , hence a canonical automorphism of order ≤ 2 of $\mathrm{Dyn}(G)$, hence also an automorphism of order ≤ 2 of $\mathrm{Of}(\mathrm{Dyn}(G))$, which we shall also denote by s_G , or simply s . Two types of parabolic subgroups $t, t' \in \mathrm{Of}(\mathrm{Dyn}(G))(S)$ will be called *opposite* when $t = s_G(t')$.

Theorem 4.3.2. *Let S be a scheme, G a reductive S -group, and P a parabolic subgroup of G .*

- (a) *If L is a Levi subgroup of P , there exists a unique parabolic subgroup P' of G such that $P \cap P' = L$.*
- (b) *For every parabolic subgroup Q of G , the following conditions are equivalent:*
 - (i) *For every $s \in S$, $((P \cap Q)_{\overline{s}})^0$ (which is smooth by 4.1.1) is reductive.*
 - (ii) *$P \cap Q$ is a Levi subgroup of P and of Q .*
 - (iii) *P and Q have opposite types, and the pair (P, Q) is in transversal position (cf. 4.2.3).*
 - (iv) *P and Q have opposite types, and $\mathrm{rad}^u(P) \cap Q = e$.*
 - (v) *$\mathrm{rad}^u(P) \cap Q = \mathrm{rad}^u(Q) \cap P = e$.*
 - (vi) *The canonical morphism $\mathrm{rad}^u(P) \times_S Q \rightarrow G$ is an open immersion.*
 - (vi') *The canonical morphism $\mathrm{rad}^u(P) \rightarrow G/Q$ is an open immersion.*

⁽²⁰⁾**N.D.E.** “Of order 2” has been replaced by “of order ≤ 2 ”, since s_G can be trivial (for example if G is of type $A_1, B_n, C_n, \dots$).

(vii) *There exist a covering family for the étale topology $\{S_i \rightarrow S\}$, and for each i a splitting (T_i, M_i, R_i) of G_{S_i} and a subset $R_i^{(1)}$ of R_i such that P_{S_i} (resp. Q_{S_i}) is the subgroup of type (R) of G_{S_i} containing T_i and defined by $R_i^{(1)}$ (resp. by $R_i^{(2)} = -R_i^{(1)}$).*

PROOF. Let us first prove the second part of the theorem; one first sees that (iii) $\Leftrightarrow$ (vii), by 4.2.1 (vii) and the definition of s_G in the split case (Exp. XXII, 3.16.2 (iv)); one evidently has (ii) $\Rightarrow$ (i); one has (vi') $\Rightarrow$ (vi) by the base change $G \rightarrow G/Q$.

Now suppose (vii) satisfied, and let us prove all the other conditions; since they are local for the étale topology, one can suppose $G = (G, T, M, R)$ split, P defined by the subset R' of R , and Q by the subset $-R'$. If L is the subgroup of type (R) of G containing T defined by $R' \cap -R'$, it is clear by Exp. XXII, 5.11.3 that L is a common Levi subgroup of P and Q . But $P = L \cdot \text{rad}^u(P)$, $Q = L \cdot \text{rad}^u(Q)$, and by Exp. XXII, 5.6.7, $\text{rad}^u(P) \cap Q = Q \cap \text{rad}^u(P) = e$; hence $P \cap Q = L$, and one has proved (ii) and (v). Since P and Q are in transversal position, the canonical morphism $P \rightarrow G/Q$ induces an open immersion $P/P \cap Q \rightarrow G/Q$ (4.2.1); but the canonical morphism $\text{rad}^u(P) \rightarrow P/P \cap Q = P/L$ is an isomorphism, so that one has proved (vi'). In view of what one has already seen, all the assertions are therefore consequences of (vii).

It now suffices for us to prove that any one of assertions (i), (iv), (v), (vi) implies (vii); since we have already proved the equivalence of (ii) and (iii), it suffices to give the proof on the geometric fibers, and one can suppose that S is the spectrum of an algebraically closed field. By 4.1.1, there exists a maximal torus T of G contained in $P \cap Q$. Let R (resp. R_1 , resp. R_2) be the set of roots of G (resp. P , resp. Q) relative to T .

Let R_1^a be the asymmetric part of R_1 (i.e. $R_1^a = \{\alpha \in R_1 \mid -\alpha \notin R_1\}$). Similarly introduce R_2^a . One must prove that $R_1 = -R_2$.

- Condition (i) implies that $R_1 \cap R_2$ is symmetric; let $\alpha \in R_1$; if $\alpha \notin R_2$, then $\alpha \in -R_2$, and if $\alpha \in R_2$, then $\alpha \in R_1 \cap R_2 = -(R_1 \cap R_2) \subset -R_2$; one therefore has $R_1 \subset -R_2$, hence, by symmetry, $R_1 = -R_2$.
- Condition (iv) implies $\text{Card}(R_1) = \text{Card}(R_2)$ and $R_1^a \cap R_2 = \emptyset$; the second condition is equivalent to $R_2 \subset -R_1$; the first then gives $R_2 = -R_1$.
- Condition (v) implies $R_1^a \cap R_2 = R_2^a \cap R_1 = \emptyset$, hence $R_2 \subset -R_1$ and $R_1 \subset -R_2$, which again gives $R_2 = -R_1$.
- Condition (vi) implies $\text{Lie}(\text{rad}^u(P)) \oplus \text{Lie}(Q) = \text{Lie}(G)$, which implies that R is the disjoint union of R_2 and R_1^a , hence that $R_2 = -R_1$.

This completes the proof of the second part of the theorem. Let us prove the first; first note that, by (vii) $\Rightarrow$ (ii), one has already proved the existence, locally for the étale topology, of the group P' sought; it therefore remains to prove its uniqueness, and this can likewise be done locally for the étale topology. One can therefore suppose G split relative to a maximal torus T of L , and P (resp. P') defined by a subset R_1 (resp. R'_1) of the root system R .

By hypothesis $R_1 \cap R'_1$ is symmetric; reasoning as above, one obtains $R'_1 = -R_1$, which proves that P' is determined by P and L , and completes the proof. $\square$

Definition 4.3.3. Two parabolic subgroups of G satisfying the equivalent conditions (i) to (vii) of 4.3.2 are called *opposite*. If P is a parabolic subgroup of G , and if L is a Levi subgroup of P (resp. and if T is a maximal torus of P), one calls the unique parabolic subgroup Q of G such that $P \cap Q = L$ (resp. such that $P \cap Q$ is the unique Levi subgroup of P containing T , cf. 1.6, or again such that $P \cap Q$ contains T and P and Q are opposite) the *parabolic subgroup opposite to P relative to L* (resp. T).

By 4.3.2 (iii), one immediately deduces from 4.2.4 and 4.2.5 parallel results; let us give a sample of them.

Corollary 4.3.4. *Let S be a scheme, G a reductive S -group, and P and Q two parabolic subgroups of G .*

- (i) *In order that P and Q be opposite, it is necessary and sufficient that, for every point $s \in S$, $P_{\bar{s}}$ and $Q_{\bar{s}}$ be opposite. If $S' \rightarrow S$ is a surjective morphism, and if $P_{S'}$ and $Q_{S'}$ are opposite, then P and Q are opposite.*
- (ii) *The functor $\underline{\text{Opp}}(G)$, such that for $S' \rightarrow S$, $\underline{\text{Opp}}(G)(S')$ is the set of pairs of opposite parabolic subgroups of $G_{S'}$, is representable by an open subscheme of $\underline{\text{Par}}(G)^2$. The functor $\underline{\text{Opp}}(/P)$, such that for $S' \rightarrow S$, $\underline{\text{Opp}}(/P)(S')$ is the set of parabolic subgroups of $G_{S'}$ opposite to $P_{S'}$, is representable by an open subscheme universally schematically dense over $S^{(21)}$ of $\underline{\text{Par}}_{s(t(P))}(G)$.*
- (iii) *Suppose P and Q opposite; let P' and Q' be two parabolic subgroups of G , P' having the same type as P . In order that P' and Q' be opposite, it is necessary and sufficient that, locally for the étale topology, the pair (P', Q') be conjugate to the pair (P, Q) . (N.B. We shall see in § 5 that one can replace the étale topology by the Zariski topology.)*

Corollary 4.3.5. *Let S be a scheme, G a reductive S -group, and P a parabolic subgroup of G .*

- (i) *The morphism $\underline{\text{Opp}}(/P) \rightarrow \underline{\text{Lev}}(P)$ (cf. 1.9), defined set-theoretically by $Q \mapsto P \cap Q$, is an isomorphism; $\underline{\text{Opp}}(/P)$ is a principal homogeneous bundle under $\text{rad}^u(P)$ ($\text{rad}^u(P)$ acting by inner automorphisms). If S is affine, there exists a parabolic subgroup of G opposite to P .*
- (ii) *Suppose S semilocal; let $\{s_i\}$ be the set of its closed points; for each i , let Q_i be a parabolic subgroup of G_{s_i} opposite to P_{s_i} . There exists a parabolic subgroup Q of G opposite to P and such that $Q_{s_i} = Q_i$ for each i .*
- (iii) *The morphism $\underline{\text{Opp}}(G) \rightarrow \underline{\text{PL}}$ (cf. 3.15), defined set-theoretically by $(P, Q) \mapsto (P, P \cap Q)$, is an isomorphism.*

All this follows from the first part of the theorem and from 1.9, 2.3 and 2.8.

Remark 4.3.6. Let P and Q be two opposite parabolic subgroups of G , and let $P \cdot Q$ be the open subscheme of G , the sheaf-theoretic image of $P \times_S Q$, introduced in 4.2.6. The “product morphism” $G \times_S G \rightarrow G$ induces isomorphisms:

$$\text{rad}^u(P) \times_S Q \xrightarrow{\sim} P \cdot Q \xleftarrow{\sim} P \times_S \text{rad}^u(Q).$$

Indeed, this follows from 4.3.2 (or from 4.2.5 (ii)) and from the fact that $P \cap Q$ is a Levi subgroup of P and of Q , hence that $P = \text{rad}^u(P) \cdot (P \cap Q)$ and $Q = \text{rad}^u(Q) \cdot (P \cap Q)$.

One similarly has a commutative diagram

$$\begin{array}{ccc} \text{rad}^u(P) & \hookrightarrow & G/Q \\ \sim \downarrow & & \downarrow \sim \\ \underline{\text{Opp}}(/P) & \hookrightarrow & \underline{\text{Par}}_{t(Q)}(G), \end{array}$$

where the vertical arrows are induced by $g \mapsto \text{int}(g)Q$.

4.4. Osculatory position

Proposition 4.4.1. *Let S be a scheme, G a reductive S -group, and P and Q two parabolic subgroups of G . The following conditions are equivalent:*

- (i) *$P \cap Q$ is a parabolic subgroup of G .*

⁽²¹⁾**N.D.E.** “Relatively dense” has been replaced by “universally schematically dense over S ”, cf. EGA IV₃, Def. 11.10.8.

- (ii) $P \cap Q$ locally contains a Borel subgroup of G for the étale topology.
- (iii) $P \cap Q$ locally contains a maximal torus of G for the étale topology, and, for every $S' \rightarrow S$ and every maximal torus T of $G_{S'}$ contained in $P_{S'}$ and $Q_{S'}$, the opposite of $P_{S'}$ relative to T is in transversal position relative to $Q_{S'}$.
- (iv) There exist a covering family for the étale topology $\{S_i \rightarrow S\}$, and for each i a maximal torus T_i of G_{S_i} contained in P_{S_i} and Q_{S_i} , and such that the opposite of P_{S_i} relative to T_i is in transversal position relative to Q_{S_i} .
- (v) There exist a covering family for the étale topology $\{S_i \rightarrow S\}$, and for each i a splitting (T_i, M_i, R_i) of G_{S_i} such that P_{S_i} (resp. Q_{S_i}) is the subgroup of type (R) of G_{S_i} containing T_i and defined by a set of roots $R_i^{(1)}$ (resp. $R_i^{(2)}$), $R_i^{(1)} \cap R_i^{(2)}$ containing a system of positive roots of R .

Moreover, if these conditions hold, one has $t(P \cap Q) = t(P) \cap t(Q)$ (with the notations of 3.2).

One has (v) $\Rightarrow$ (ii) and (iii) $\Rightarrow$ (iv) trivially. Moreover, (ii) $\Rightarrow$ (i) by 1.18. One has (iv) $\Rightarrow$ (v): indeed, one can suppose G split, P (resp. Q) defined by the set of roots R_1 (resp. R_2); the opposite of P is then defined by $-R_1$, and one is reduced to lemma 4.2.2. One proves (i) $\Rightarrow$ (iii) by splitting, in the same way. Finally, the last assertion of the theorem can be proved locally for the étale topology; one can suppose that $P \cap Q$ contains a Borel subgroup B of G , and one is reduced to 3.7.

Definition 4.4.2. Two parabolic subgroups of G satisfying conditions (i) to (v) of 4.4.1 are said to be *in osculatory position*.

Corollary 4.4.3. Let P and Q be two parabolic subgroups in osculatory position, and let P' and Q' be two parabolic subgroups of G of the same types as P and Q respectively; in order that P' and Q' be in osculatory position, it is necessary and sufficient that the pair (P', Q') be conjugate to the pair (P, Q) , locally for the étale topology.

It suffices to prove that, if P and P' are in osculatory position relative to Q , they are conjugate, locally for (ét), by a section of Q . Now $P \cap Q$ and $P' \cap Q$ are two parabolic subgroups of the same type contained in Q , and so are conjugate, locally for (ét), by a section of Q , by part (ii) of the lemma below. One can therefore suppose $P \cap Q = P' \cap Q$; one then has $P = P'$, by part (i) of the same lemma:

Lemma 4.4.4. Let P, P' and Q be three parabolic subgroups of the reductive S -group G .

- (i) In order that $P = P'$, it is necessary and sufficient that P and P' be in osculatory position and of the same type.
- (ii) If $P \subset Q$, $P' \subset Q$, and if $g \in G(S)$ is such that $\text{int}(g)P$ and P' are in osculatory position, then $g \in Q(S)$.

Part (i) follows trivially from the last assertion of 4.4.1. Let us prove (ii): Q and $\text{int}(g)Q$ contain $P' \cap \text{int}(g)P$, and so are in osculatory position; they coincide by (i), hence $g \in \underline{\text{Norm}}_G(Q)(S) = Q(S)$.

Note that assertions (iii) and (iv) of the theorem immediately give:

Corollary 4.4.5. Let P, P' and Q be three parabolic subgroups of the reductive S -group G containing the same maximal torus T of G . Suppose P and P' opposite relative to T . In order that Q be in osculatory position relative to P , it is necessary and sufficient that it be in transversal position relative to P' . Under these conditions, $P \cap Q$ is also in transversal position relative to P' .

Corollary 4.4.6. Let P and Q be two parabolic subgroups of G containing the same maximal torus T . In order that P and Q be in transversal position, it is necessary and sufficient that

there exist two parabolic subgroups $P' \subset P$ and $Q' \subset Q$ of G opposite relative to T ; one can even choose $t(P') = t(P) \cap s(t(Q))$.⁽²²⁾

The condition is evidently sufficient (4.2.1 (i) and 4.3.2 (iii)). Let us show that it is necessary; let P^- (resp. Q^-) be the opposite of P (resp. Q) relative to T . By 4.4.5, $P^- \cap Q$ is in transversal position relative to P and Q^- , hence also relative to $P \cap Q^-$ by a new application of 4.4.5; moreover

$$\begin{aligned} t(P^- \cap Q) &= t(P^-) \cap t(Q) = s(t(P)) \cap s(t(Q^-)) \\ &= s(t(P) \cap t(Q^-)) = s(t(P \cap Q^-)), \end{aligned}$$

hence $P^- \cap Q = P'$ and $P \cap Q^- = Q'$ are opposite (4.3.2 (iii)); but $P' \cap Q' \supset T$, so they are indeed opposite relative to T .

4.5. Standard position

In this section, we briefly indicate how some of the preceding results generalize.

Proposition 4.5.1. *If P_1 and P_2 are two parabolic subgroups of the reductive S -group G , the following conditions are equivalent.*⁽²³⁾

- (i) $P_1 \cap P_2$ is smooth.
- (ii) $P_1 \cap P_2$ is a subgroup of type (R) (or of type (RC)) of G .
- (iii) $P_1 \cap P_2$ locally contains a maximal torus of G for the (fpqc) topology.
- (iv) $P_1 \cap P_2$ locally contains a maximal torus of G for the Zariski topology.

When S is semilocal, these conditions are moreover equivalent to:

- (v) $P_1 \cap P_2$ contains a maximal torus of G .

PROOF. ⁽²³⁾ Evidently, (iv) $\Rightarrow$ (iii) (and (v) $\Rightarrow$ (iv) when S is semilocal), and (ii) $\Rightarrow$ (i) by Exp. XXII, Def. 5.2.1. We shall show (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii), then that (iii) implies (i) and (iv) (and also (v) when S is semilocal). Put $K = P_1 \cap P_2$. By 4.1.1 and [BT65], 4.5, each geometric fiber $K_{\bar{s}}$ contains a maximal torus $T_{\bar{s}}$ of $G_{\bar{s}}$ and is connected, and the set of roots of $K_{\bar{s}}$ relative to $T_{\bar{s}}$ is a closed subset of the set of roots of $G_{\bar{s}}$ relative to $T_{\bar{s}}$. Thus, if K is smooth over S , it is a subgroup of type (RC) (hence a fortiori of type (R)), cf. Exp. XXII 5.2.1 and 5.11.1. One therefore has (i) $\Leftrightarrow$ (ii).

If K is a subgroup of type (R), it locally contains a maximal torus of G for the étale topology, by Exp. XXII 2.2 and Exp. XIX 6.1. Thus (ii) $\Rightarrow$ (iii).

Suppose (iii) satisfied, and let us show that K is smooth. By (fpqc) descent, one can suppose that $G = (G, T, M, R)$ is split, where T is a maximal torus contained in $P \cap Q$, and that there exist two closed subsets R_1 and R_2 of R such that $P = H_{R_1}$ and $Q = H_{R_2}$.

Since K has connected fibers, it follows from Exp. XXII 5.4.5 that K equals $H_{R_1 \cap R_2}$, and so is smooth over S and of type (RC).

Let R_1^s be the symmetric part of R_1 , and $R_1^a = R_1 - R_1^s$; then $\text{rad}^u(P_1) = H_{R_1^a}$. As noted in the proof of [BT65], 4.4, $R' = (R_1^s \cap R_2) \cup R_1^a$ is a closed subset of R such that $R' \cup (-R') = R$; thus, by Exp. XXI 3.3.6, R' contains a system of positive roots of R . Moreover, the symmetric part of R' is $R_1^s \cap R_2^s$, which is contained in $R_1 \cap R_2$.

One deduces from the foregoing that $P' = K \cdot \text{rad}^u(P)$ is a parabolic subgroup of G , and that K is a subgroup of type (RC) of P' such that $\text{rad}^u(K) = K \cap \text{rad}^u(P')$. Thus, by 2.11, K possesses a Levi subgroup L . It then follows from Exp. XIV 3.20 and 3.21 that K satisfies assertion (iv), as well as assertion (v) when S is semilocal. This proves 4.5.1. $\square$

⁽²²⁾N.D.E. Recall that the involution s was defined in 4.3.1.

⁽²³⁾N.D.E. The proof of the equivalence of these conditions has been added (as has condition (v), used implicitly in 6.17 of the original); consequently no. 4.5.1 has been changed into proposition 4.5.1 plus definition 4.5.1.1.

Definition 4.5.1.1. ⁽²⁴⁾ When the preceding conditions hold, one says that P and Q are in *mutual standard position*; this is the case, for example, if P and Q are in transversal position, or in osculatory position, or if the base is the spectrum of a field. This notion is stable under extension of the base and local for the (fpqc) topology.

4.5.2. — Let (P, Q) and (P', Q') be two pairs of parabolic subgroups of G in standard position, and let $\underline{H}$ be the subfunctor of G defined as follows: $\underline{H}(S')$ is the set of $g \in G(S')$ such that $\text{int}(g)P = P'$ and $\text{int}(g)Q = Q'$. This is a closed subscheme of G , smooth over S , and formally principal homogeneous under $P \cap Q$. One deduces that the following conditions are equivalent:

- (i) (P, Q) and (P', Q') are locally conjugate for the (fpqc) topology,
- (ii) (P, Q) and (P', Q') are locally conjugate for the étale topology.
- (iii) (P, Q) and (P', Q') are conjugate on every geometric fiber.

One then says that the pairs (P, Q) and (P', Q') have the *same type of mutual position*. This notion is stable under base change and local for the (fpqc) topology.

4.5.3. — Let $\underline{\text{Stand}}(G)$ be the subfunctor of $\underline{\text{Par}}(G) \times_S \underline{\text{Par}}(G)$ “consisting of pairs in mutual standard position”. Then $\underline{\text{Stand}}(G)$ is representable; there exist a finite étale S -scheme $\underline{\text{TypeStand}}$ (“scheme of types of mutual standard position”) and a smooth morphism of finite presentation, with irreducible geometric fibers (and hence in particular faithfully flat),

$$t_2 : \underline{\text{Stand}}(G) \longrightarrow \underline{\text{TypeStand}},$$

which is a quotient of $\underline{\text{Stand}}(G)$ by the action of G : two sections of $\underline{\text{Stand}}(G)$ (over any $S' \rightarrow S$) have the same type of mutual position if and only if they have the same image under t_2 .

One has a commutative diagram

$$\begin{array}{ccc} \underline{\text{Stand}}(G) & \xrightarrow{t_2} & \underline{\text{TypeStand}} \\ \downarrow & & \downarrow q \\ \underline{\text{Par}}(G) \times_S \underline{\text{Par}}(G) & \xrightarrow{t \times t} & \underline{\text{Of}}(\underline{\text{Dyn}}(G)) \times_S \underline{\text{Of}}(\underline{\text{Dyn}}(G)) \end{array}$$

where the morphism q can be described by descent as follows: if (P, Q) is a pair of parabolic subgroups of G in standard relative position, and if T is a maximal torus of $P \cap Q$, then the morphism $\underline{\text{Norm}}_G(T) \rightarrow \underline{\text{Stand}}(G)$ defined set-theoretically by $n \mapsto (P, \text{int}(n)Q)$ induces an isomorphism

$$W_P(T) \backslash W_G(T) / W_Q(T) \simeq q^{-1}(t(P), t(Q)).$$

(The first side denotes the sheaf of double cosets. . . .) These assertions are proved without difficulty (note in particular that $t_2^{-1}(t_2(P, Q)) \simeq G/(P \cap Q)$).

4.5.4. — Now let P be a fixed parabolic subgroup of G , and let $\underline{\text{Par}}(G; P)$ be the functor of parabolic subgroups of G in standard position relative to P . For each $t \in \underline{\text{Of}}(\underline{\text{Dyn}}(G))(S)$, similarly put $\underline{\text{Par}}_t(G; P) = \underline{\text{Par}}(G; P) \cap \underline{\text{Par}}_t(G)$. One sees immediately that the preceding two functors are obtained from $\underline{\text{Stand}}(G)$ by fibered products, and so are representable by S -schemes smooth and of finite presentation over S , with nonempty fibers. One has a canonical morphism t_P induced by t_2 (i.e. $t_P(Q) = t_2(P, Q)$)

$$t_P : \underline{\text{Par}}_t(G; P) \longrightarrow q^{-1}(t(P), t)$$

which is smooth and of finite presentation, with irreducible geometric fibers. The canonical morphism $\underline{\text{Par}}_t(G; P) \rightarrow \underline{\text{Par}}_t(G)$ is a surjective monomorphism, and can therefore be

⁽²⁴⁾**N.D.E.** See N.D.E. (23). Moreover, for an example of parabolic subgroups P, Q which are not in standard position, see 7.11 below.

considered as a *cellular decomposition* of $\underline{\text{Par}}_t(G)$ (indexed by the set of connected components of $q^{-1}(t(P), t)$).

4.5.5. — Now suppose that the type t is of the form $t(Q)$, where Q is a parabolic subgroup of G in standard position relative to P , and that $P \cap Q$ contains a maximal torus T .

Then $\underline{\text{Par}}_t(G) \simeq G/Q$ and $q^{-1}(t(P), t) \simeq W_P(T) \backslash W_G(T) / W_Q(T)$, which gives a diagram

$$\begin{array}{ccc} \underline{\text{Par}}_{t(Q)}(G; P) & \xrightarrow{f} & W_P(T) \backslash W_G(T) / W_Q(T) \\ \downarrow i & & \\ \underline{\text{Par}}_{t(Q)}(G) & \xrightarrow{\sim} & G/Q \end{array}$$

where i is a surjective monomorphism, and where f is smooth and of finite presentation, with irreducible geometric fibers. Moreover, if Q_1 and Q_2 are two sections of $\underline{\text{Par}}_{t(Q)}(G; P)$ (over an $S' \rightarrow S$), that is, two parabolic subgroups of $G_{S'}$ conjugate (locally for (fpqc)) to Q and in standard position relative to $P_{S'}$, then Q_1 and Q_2 are conjugate by a section of P (locally for (fpqc)) if and only if $f(Q_1) = f(Q_2)$. If S is the spectrum of an algebraically closed field k , one thus finds the relation

$$P(k) \backslash G(k) / Q(k) \simeq W_P(T)(k) \backslash W_G(T)(k) / W_Q(T)(k).$$

In general, if one assumes that the scheme $W_P(T) \backslash W_G(T) / W_Q(T)$ is constant and of the form E_S (which is the case, for example, when G is split relative to T and S is connected), the $f^{-1}(e)$, $e \in E$, form a decomposition of $\underline{\text{Par}}_{t(Q)}(G; P)$ into open and closed subschemes, which are homogeneous spaces under P , smooth and of finite presentation over S , with irreducible geometric fibers.

4.5.6. — Let us return to the general situation of 4.5.4. The scheme $q^{-1}(t(P), t)^{(25)}$ always has two particular sections, corresponding respectively to the types “transversal position” and “osculatory position”. The inverse image of the first section is a relatively dense open subscheme of $\underline{\text{Par}}_t(G; P)$, as one has seen above; it is the cell of maximum relative dimension of the decomposition.

The inverse image of the second section is presumably a closed subscheme of $\underline{\text{Par}}_t(G; P)$; it is the cell of minimum relative dimension of the decomposition.

5. Conjugacy theorem

Theorem 5.1. *Let S be a semi-local scheme, G a reductive S -group, P and P' two opposite parabolic subgroups (4.3.3) of G . Then*

$$\text{rad}^u(P)(S) P' \cdot P = G,$$

i.e. the union of the open subschemes $uP' \cdot P$ (4.3.6), as u ranges over $\text{rad}^u(P)(S)$, is the whole of G .

The proof proceeds in several steps:

5.1.1. — It suffices to give the proof in the case where S is the spectrum of a field k ; this follows at once from 2.6.

5.1.2. — Let $L = P \cap P'$. Suppose that L has a Borel subgroup B_L ; let T be a maximal torus of B_L (Exp. XXII, 5.9.7); one checks at once that $B = B_L \cdot \text{rad}^u(P)$ is a Borel subgroup of P ; let B' be the Borel subgroup of G opposite to B relative to T (i.e. such that $B \cap B' = T$). One has $B' \subset P'$, as one checks at once by splitting G relative to T . Let us prove that

$$(x) \quad B^u(S) B' \cdot B \subset \text{rad}^u(P)(S) P' \cdot P.$$

⁽²⁵⁾**N.D.E.** One has corrected $t^{-1}(t(P), t)$ to $q^{-1}(t(P), t)$ and, below, $\underline{\text{Par}}_t(G)$ to $\underline{\text{Par}}_t(G; P)$ (twice).

Since one has

$$B^u(S) \subset P(S) = \text{rad}^u(P)(S) \cdot L(S) \subset \text{rad}^u(P)(S)P'(S),$$

it suffices to prove that $B' \cdot B \subset P' \cdot P$, which is obvious. It follows from (x) that it suffices to prove 5.1 for the pair (B, B') .

5.1.3. — The theorem is true if k is algebraically closed; indeed, the condition of 5.1.2 is satisfied, and one concludes by Exp. XXII, 5.7.10.

5.1.4. — The theorem is true when k is an infinite field. Indeed, $\text{rad}^u(P)(k)$ is dense in $\text{rad}^u(P)(\bar{k})$ by 2.7, and the theorem is true for $\bar{k}$.

5.1.5. — One is therefore reduced to the case where k is a finite field. Now $\text{Bor}(L)$ is a *smooth* homogeneous space of L ; it therefore follows from Lang's theorem (Am. J. of Maths., 78, 1956⁽²⁶⁾) that L has a Borel subgroup B_L . By 5.1.2, one can therefore assume that $P = B$ and $P' = B'$ are Borel subgroups. One writes $T = B \cap B'$.

5.1.6. — Let K be the algebraic closure of k ; choose a pinning of the triple (G_K, B_K, T_K) ; let R_+ (resp. Δ) be the set of positive (resp. simple) roots. By Exp. XXII, 5.7.2, it suffices to prove that for every $\alpha \in \Delta$, one has

$$(1) \quad u_\alpha B'^u(K) \subset B^u(k) B'^u(K) B(K).$$

Let α_i be the different roots conjugate to α over k (they are elements of Δ , since B is “defined over k ”), and let R' be the set of roots that are linear combinations of the α_i . Write $R'_- = R_- \cap R'$. Since “ R' is defined over k ”, there is a subtorus Q of T such that Q_K is the maximal torus of the common kernel of the α_i .

Write $Z = \text{Centr}_G(Q)$, $B_Z = B \cap Z$, $B'_Z = B' \cap Z$ (cf. Exp. XXII, 5.10.2). Let us show that it suffices to verify the desired assertion in Z , that is,

$$(2) \quad u_\alpha \cdot B'_Z{}^u(K) \subset B_Z^u(k) B'_Z{}^u(K) B_Z(K).$$

One has $(B_Z^u)_K = \prod_{\alpha \in R'_-} U_\alpha$; let R'' be the complement of R' in R , and put

$$V = \prod_{\alpha \in R'' \cap R_-} U_\alpha.$$

One has at once $(B'^u)_K = (B_Z^u)_K \cdot V$, and $(B_Z)_K$ normalizes V (Exp. XXII, 5.6.7). One therefore deduces successively from (2)

$$\begin{aligned} u_\alpha \cdot B'^u(K) &= u_\alpha \cdot B_Z^u(K) V(K) \\ &\subset B_Z^u(k) B_Z^u(K) B_Z(K) V(K) \\ &\subset B_Z^u(k) B_Z^u(K) V(K) B_Z(K), \end{aligned}$$

which implies (1) at once.

We are therefore reduced to the case where $G = Z$, that is, where the Galois group of K over k acts transitively on the simple roots.

5.1.7. — The assertion to be proved is equivalent to the fact that G/B is the union of the translates by $B^u(k)$ of the open image of B'^u , an assertion that does not change if one replaces G by its adjoint group (or, for that matter, by any group giving the same adjoint group). One can therefore assume that G is adjoint.

5.1.8. — Consider then the Dynkin diagram of G_K . The Galois group acts *transitively* on this Dynkin diagram. But this Galois group has only cyclic quotients, and the Dynkin

⁽²⁶⁾N.D.E. See also [DG70], § III.5, 7.4.

diagram has no cycles. It follows at once that this diagram is of type nA_1 , $n \geq 0$, or mA_2 , $m \geq 0$. Using the canonical decomposition of Exp. XXIV, 5.9, one can write

$$G = \prod_{D/K} G_0,$$

where D is a *finite* K -scheme and G_0 is either a torus, or of type A_1 , or of type A_2 .

By Exp. XXIV, 5.12, B comes from a Borel subgroup B_0 of G_0 , T from a maximal torus T_0 of G_0 ; B' comes from the Borel subgroup B'_0 of G_0 opposite to B_0 relative to T_0 . One has

$$B'^u(k) = B_0^u(D), \quad B'^u \cdot T \cdot B^u = \prod_{D/k} B_0'^u \cdot T_0 \cdot B_0^u,$$

and it suffices for us to prove the desired assertion for the triple (G_0, B_0, T_0) .

5.1.9. — One can therefore assume that G is of type $\emptyset$, A_1 or A_2 . Since G has a Borel subgroup B , G is quasi-splittable relative to B (Exp. XXIV, 3.9.1), hence splittable if it is of type $\emptyset$ or A_1 . Since the theorem has already been proved in the split case (Exp. XXII 5.7.10), only the case A_2 remains to be treated. By Exp. XXIV, 3.11 there is a morphism $E \rightarrow \text{Spec}(k)$, a Galois principal bundle under the group $\mathbf{Z}/2\mathbf{Z}$ of automorphisms of the Dynkin diagram of type A_2 , such that $G = \text{Q-}\dot{\text{E}}\text{p}_{E/\text{Spec}(k)}(A_2)$. If E has a section, G is splittable and the theorem is proved. Otherwise, one necessarily has $E = \text{Spec}(k')$, where k' is a quadratic extension of k . Finally, as one has seen in 5.1.7, one can assume that G is simply connected (i.e. that G is a form of $\text{SL}_{3,k}$).

5.1.10. — One is therefore in the following situation: one has a finite field k , a quadratic extension k' of k . The group $\text{SL}_{3,k'}$ of 3×3 matrices of determinant 1 is pinned as follows: the maximal torus is the group of diagonal matrices, the Borel subgroup is the group of upper triangular matrices, the “pins” are the elements:

$$u_\alpha = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad u_\beta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

One sees at once that the big cell Ω is defined by

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \in \Omega(S) \iff a \text{ and } ae - bd \text{ are invertible,}$$

and that

$$B^u(k) = \left\{ \begin{pmatrix} 1 & \bar{x} & z \\ 0 & 1 & x \\ 0 & 0 & 1 \end{pmatrix} \middle| x, z \in k', \quad z + \bar{z} = x\bar{x} \right\}.$$

We must prove inclusion (1) of 5.1.6, that is, show that for all $a, b, c \in K$ (algebraic closure of k), there exist $x, z \in k'$ such that $z + \bar{z} = x\bar{x}$ and

$$(1) \quad \begin{pmatrix} 1 & \bar{x} & z \\ 0 & 1 & x \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{pmatrix} \in \Omega(K).$$

- If $a \neq -1$, one takes $x = z = 0$.
- If $a = -1$, the conditions to be satisfied read

$$\begin{cases} z + \bar{z} = x\bar{x}, \\ bz - \bar{x} \neq 0, \\ (b + c)\bar{z} + bx - 1 \neq 0. \end{cases}$$

Let q be the number of elements of k ($q \geq 2$). One knows that for every $m \in k$, the equation $z + \bar{z} = m$, with $z \in k'$, has q solutions.

- If $b = 0$, take $x = 1$; one must solve $z + \bar{z} = 1$, $c\bar{z} \neq 1$, which is always possible by the preceding remark.
- If $b \neq 0$, take $x = 0$; one must solve

$$z + \bar{z} = 0, \quad z \neq 0, \quad (b + c)z \neq 1.$$

This is always possible if $q \geq 3$. If $k = \mathbf{F}_2$, it is possible if $b + c \neq 1$; one can take $z = 1$.

- Only the case $k = \mathbf{F}_2$, $b + c = 1$, $b \neq 0$ therefore remains to be treated. The system then reads

$$z + \bar{z} = x\bar{x}, \quad bz \neq \bar{x}, \quad \bar{z} + bx \neq 1.$$

If $b = 1$ (resp. $b \notin k'$), put $x = 1$; then the last two conditions read $z \neq b^{-1}, 1 - b$, and they follow from $z + \bar{z} = 1$, which has solutions. Finally, if $b \in k' - k$, one can take $x = \bar{b}$, $z = b$. Q.E.D.

Corollary 5.2. *Let S be a semi-local scheme, G a reductive S -group, P and P' two opposite parabolic subgroups of G . The canonical map*

$$\mathrm{rad}^u(P)(S) \cdot \mathrm{rad}^u(P')(S) \longrightarrow (G/P)(S)$$

is surjective (in particular, one has $(G/P)(S) = G(S)/P(S)$). Every parabolic subgroup Q of G of the same type as P is of the form $\mathrm{int}(uu')P$, with $u \in \mathrm{rad}^u(P)(S)$ and $u' \in \mathrm{rad}^u(P')(S)$.

The second assertion is obviously equivalent to the first; let us prove the latter. Let s_i be the closed points of S , let V be the open subscheme of G/P that is the image of P' (and is isomorphic to $\mathrm{rad}^u(P')$, cf. 4.3.6), and let $x \in G/P(S)$. By 5.1, for each i there is a section $u_i \in \mathrm{rad}^u(P)(\kappa(s_i))$ such that $u_i x_{s_i}$ is a section of V_{s_i} . If $u \in \mathrm{rad}^u(P)(S)$ lifts the u_i (2.6), ux is a section of V , since such an assertion is checked on the closed fibers. But $\mathrm{rad}^u(P')(S) \xrightarrow{\sim} V(S)$ is bijective, and one concludes at once.

Corollary 5.3. *Let S be a semi-local scheme, G a reductive S -group, P , P' and Q three parabolic subgroups of G . There is $g \in G(S)$ such that $\mathrm{int}(g)Q$ is in transversal position relative to P and P' .*

With the notation of 4.2.4(ii), one must verify that the open subscheme of G that is universally schematically dense over $S^{(27)}$, $\underline{\mathrm{Gen}}(Q/P) \cap \underline{\mathrm{Gen}}(Q/P')$, has a section over S . In fact, choose a parabolic subgroup of G opposite to Q (4.3.5(i)), say Q_1 , and put $U = \mathrm{rad}^u(Q)$, $U' = \mathrm{rad}^u(Q_1)$. We shall show that there is $g \in U(S)U'(S)$ answering the question; in this form it follows from 4.2.4(i) and 2.6 that it suffices to verify the assertion on the fibers at the closed points of S , and one can therefore assume that S is the spectrum of a field k .

If k is algebraically closed, there is $g \in G(k)$ answering the question; now g is written $uu'q$, with $u \in U(k)$, $u' \in U'(k)$, $q \in Q(k)$ (5.2), and one has $\mathrm{int}(uu')Q = \mathrm{int}(g)Q$.

If k is infinite, consider the open subscheme V of $U \times_k U'$ defined by the cartesian diagram

$$\begin{array}{ccc} U \times_k U' & \longrightarrow & G \\ \uparrow & & \downarrow \\ V & \longrightarrow & \underline{\mathrm{Gen}}(Q/P) \cap \underline{\mathrm{Gen}}(Q/P') \end{array}$$

since $V(\bar{k}) \neq \emptyset$ by what one has just seen, V is dense in $U \times_k U'$, hence has a section by 2.7.

If k is finite, P (resp. P') has a Borel subgroup B (resp. B'), by Lang's theorem (cf. 5.1.5), the schemes $\underline{\mathrm{Bor}}(P) \simeq \underline{\mathrm{Bor}}(P/\mathrm{rad}^u(P))$ and $\underline{\mathrm{Bor}}(Q) \simeq \underline{\mathrm{Bor}}(Q/\mathrm{rad}^u(Q))$ being smooth. If B_1 is a Borel subgroup opposite to B (4.3.5(i)), there are $a \in B^u(k)$ and $a_1 \in B_1^u(k)$ such that $\mathrm{int}(aa_1)B = B'$ (5.2); then $B_0 = \mathrm{int}(aa_1)B_1 = \mathrm{int}(a)B_1$ is opposite to B' and to B ; if Q_0 is the unique parabolic subgroup of G containing B_0 and of the same type as Q (3.8), Q_0

⁽²⁷⁾**N.D.E.** One has replaced “relatively dense” by “universally schematically dense over S ”, cf. EGA IV₃, Def. 11.10.8.

is in transversal position relative to P and P' (4.2.1(vi)). Moreover, by 5.2, Q_0 is written $\text{int}(uu')Q$, with $u' \in U'(k)$, $u \in U(k)$, which was to be proved.

Corollary 5.4. *Let S be a semi-local scheme, G a reductive S -group, P and Q two parabolic subgroups of G . There is $g \in G(S)$ such that $\text{int}(g)P$ is in osculatory position relative to Q , i.e. (4.4.2) such that $\text{int}(g)P \cap Q$ is a parabolic subgroup of G .*

Indeed, by 4.3.5(i), there is a parabolic subgroup P' of G opposite to P . By 5.3, there is a parabolic subgroup P'_1 of G of the same type as P' , in transversal position relative to P and Q . If T is a maximal torus of $P'_1 \cap Q$ (4.2.7(ii)), and if P_1 is the opposite of P'_1 relative to T , then P_1 and Q are in osculatory position, by 4.4.5. Moreover, since P and P_1 are opposite to P'_1 , there is $g \in \text{rad}^u(P'_1)(S)$ such that $\text{int}(g)P = P_1$ (4.3.5(i)). Q.E.D.

Observe, moreover, that for the same reason, there is $u \in \text{rad}^u(P)(S)$ such that $\text{int}(u)P' = P'_1$, hence g is written $\text{int}(u)u'$, with $u' \in \text{rad}^u(P')(S)$, which gives $P_1 = \text{int}(uu'u^{-1})P = \text{int}(uu')P$ and, in passing, proves 5.2 again.

Statements 5.3 and 5.4 are the essential results of this paragraph. Let us first state a few consequences of 5.4.

Corollary 5.5. *Let S be a semi-local scheme, G a reductive S -group.*

(i) *If P and Q are two parabolic subgroups of G and if $t(P) \subset t(Q)$ (cf. 3.3), there is $g \in G(S)$ such that $\text{int}(g)P \subset Q$.*

(ii) *Let*

$$P_1 \supset P_2 \supset \cdots \supset P_n \quad \text{and} \quad P'_1 \supset P'_2 \supset \cdots \supset P'_n$$

be two chains of parabolic subgroups of G such that $t(P_i) = t(P'_i)$. There is $g \in G(S)$ such that $\text{int}(g)P_i = P'_i$ for every i .

(iii) *Let P, Q, P', Q' be four parabolic subgroups of G such that $t(P) = t(P')$ and $t(Q) = t(Q')$. If the pairs (P, P') and (Q, Q') are in transversal (resp. osculatory) position, there is $g \in G(S)$ such that $\text{int}(g)P = P'$ and $\text{int}(g)Q = Q'$.*

(iv) *Let P and P' be two parabolic subgroups of the same type, L (resp. L') a Levi subgroup of P (resp. P'). There is $g \in G(S)$ such that $\text{int}(g)P = P'$ and $\text{int}(g)L = L'$.*

PROOF. (i) follows at once from 5.4; (ii) is proved by induction on n , the case $n = 0$ being trivial; one can therefore assume that $P_i = P'_i$ for $i = 1, \dots, n-1$; by 5.2 there is $g \in G(S)$ such that $\text{int}(g)P_n = P'_n$; but then P_n and $\text{int}(g)P_n$ are contained in $P_{n-1} = P'_{n-1}$, hence $g \in P_{n-1}(S)$ (4.4.4(ii)) and $\text{int}(g)P_i = P'_i$ for $i = 1, \dots, n$.

Moreover, (iv) follows at once from 5.2 and 1.8. Let us prove (iii) in the case “transversal position”; the assertion follows from (iv) when the types of P and Q are opposite (4.3.3(iii)); in the general case, by 4.2.7(iii) and 4.4.6 one can find parabolic subgroups P_1, P'_1, Q_1, Q'_1 of P, P', Q, Q' respectively, such that P_1 and P'_1 are opposite, as are Q_1 and Q'_1 , and such that $t(P_1) = t(P'_1)$; there is therefore $g \in G(S)$ such that $\text{int}(g)P_1 = P'_1$, $\text{int}(g)Q_1 = Q'_1$, and one can assume that $P_1 = P'_1$ and $Q_1 = Q'_1$; but then P and P' are in osculatory position and of the same type, hence $P = P'$ (4.4.4(i)); for the same reason $Q = Q'$.

It remains for us to prove assertion (iii) in the case “osculatory position”. By the conjugacy theorem (5.2), one can assume that $P = P'$; by the same theorem, one can find $g \in G(S)$ such that $\text{int}(g)Q = Q'$; but then $g \in P(S)$ by 4.4.4(ii), and one has $\text{int}(g)P = P = P'$. $\square$

Definition 5.6. Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G . One says that P is *minimal* if whenever Q is a parabolic subgroup of G contained in P , one has $Q = P$.

One will note that this notion is not, in general, stable under passage to fibers.

Corollary 5.7. *Let S be a semi-local scheme, G a reductive S -group.*

(i) *Let $t, t' \in \underline{\text{Of}}(\underline{\text{Dyn}}(G))(S)$. If there exist in G a parabolic subgroup of type t and a parabolic subgroup of type t' , there exists a parabolic subgroup of type $t \cap t'$. In*

- particular, there is a least element $t_{\min}$ in the set of the $t(P)$, as P ranges over the set of parabolic subgroups of G .
- (ii) Every parabolic subgroup of G contains a minimal parabolic subgroup. For a parabolic subgroup of G to be minimal, it is necessary and sufficient that it be of type $t_{\min}$. Two minimal parabolic subgroups of G are conjugate by an element of $G(S)$.

This follows at once from 5.4 and 5.5(i).

Remark 5.8. A parabolic subgroup opposite to a minimal parabolic subgroup is also minimal; this implies $s(t_{\min}) = t_{\min}$.

Corollary 5.9. Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G . The canonical morphism

$$G \longrightarrow G/P = X$$

makes G a locally trivial X -bundle (in the Zariski sense) with group P_X . If L is a Levi subgroup of P , the canonical morphism (cf. 3.12)

$$G \longrightarrow G/L = Y$$

makes G a locally trivial Y -bundle (in the Zariski sense) with group L_Y .

It suffices to prove that if one has a morphism $S' \longrightarrow S$, where S' is local, and a morphism $S' \longrightarrow X$ (resp. $S' \longrightarrow Y$), it lifts to a morphism $S' \longrightarrow G$. In other words, one can assume that S is local and must show that the map $G(S) \longrightarrow X(S)$ (resp. $G(S) \longrightarrow Y(S)$) is surjective.

The first assertion was proved in 5.2; let us prove the second. Let $y \in Y(S)$; its canonical image in $X(S)$ comes from a $g \in G(S)$; the projection y' of g in $Y(S)$ therefore has the same projection as y in $X(S)$. There is therefore a unique $u \in \text{rad}^u(P)(S)$ such that $y'u = y$, and the projection of gu in $Y(S)$ is indeed y .

Corollary 5.10. Let S be a semi-local scheme, G a reductive S -group.

- (i) Let P be a parabolic subgroup of G , L a Levi subgroup of P . The canonical maps (cf. 3.21) induce bijections

$$H^1(S, L) \xrightarrow{\sim} H^1(S, P) \xrightarrow{\sim} H^1_{t(P)}(S, G).$$

- (ii) Let $t, t' \in \underline{\text{Of}}(\underline{\text{Dyn}}(G))(S)$; one has (cf. 3.21)

$$H^1_t(S, G) \cap H^1_{t'}(S, G) = H^1_{t \cap t'}(S, G).$$

- (iii) If P and Q are two parabolic subgroups of G in osculatory position, the following canonical diagram is cartesian and consists of injections:

$$\begin{array}{ccc} H^1(S, P) & \longrightarrow & H^1(S, G) \\ \uparrow & & \uparrow \\ H^1(S, P \cap Q) & \longrightarrow & H^1(S, Q) \end{array}$$

Let us prove (i). The map $H^1(S, L) \longrightarrow H^1(S, P)$ is bijective by 2.3; the map $H^1(S, P) \longrightarrow H^1_{t(P)}(S, G)$ is surjective (3.21); let us show that it is injective, i.e. that the canonical map $H^1(S, P) \longrightarrow H^1(S, G)$ is injective. Let Q be a principal bundle under P , Q_1 the associated principal bundle under G , P' and G' the corresponding twisted forms of P and G . It is clear that G' is a reductive S -group and that P' is a parabolic subgroup of it. The set of elements of $H^1(S, P)$ that have the same image as the class of Q in $H^1(S, G)$ is naturally identified with the kernel of the canonical map $H^1(S, P') \longrightarrow H^1(S, G')$, and the latter, by the cohomology exact sequence, with the set of orbits of $G'(S)$ in $(G'/P')(S)$ (for these

arguments in nonabelian cohomology, see Giraud's thesis⁽²⁸⁾). But $G'(S)$ acts transitively on $(G'/P')(S)$ by 5.2.

Let us prove (ii): let Q be a principal homogeneous bundle under G and G^Q the corresponding twisted form of G . By definition (3.21), we must prove that G^Q has a parabolic subgroup of type $t \cap t'$ if and only if it has parabolic subgroups of type t and t' , which is simply the conjunction of 3.8 and 5.7(i). Finally, (iii) follows at once from (i) and (ii).

Let us now state a consequence of 5.3.

Corollary 5.11. *Let S be a semi-local scheme, G a reductive S -group, P a parabolic subgroup of G . If $P \neq G$, there are at least three distinct parabolic subgroups of G of the same type as P ; in other words, $P \neq G$ implies $(G(S) : P(S)) \geq 3$.*

Indeed, let P' be a parabolic subgroup of G opposite to P (4.3.5(i)). Since $P \neq G$, one has $\text{rad}^u(P') \neq e$ (by 4.3.2, for example). By 2.1, $\text{rad}^u(P')(S) \neq e$; let therefore $u \in \text{rad}^u(P')(S)$, $u \neq e$. Then $\text{int}(u)P \neq P$, and by 5.3 there is a P_1 of the same type as P and opposite to P and $\text{int}(u)P$; then P_1, P and $\text{int}(u)P$ are three distinct parabolic subgroups of G of the same type as P .

6. Parabolic subgroups and split tori

Proposition 6.1. *Let S be a scheme, G a reductive S -group, $\mathfrak{g} = \text{Lie}(G/S)$, and Q a split subtorus of G . Write $Q = D_S(M)$ and let*

$$\mathfrak{g} = \coprod_{\alpha \in M} \mathfrak{g}^\alpha$$

be the decomposition of $\mathfrak{g}$ under the action of Q . Let M_1 be a subset of M such that $0 \in M_1$ and $\alpha, \beta \in M_1 \Rightarrow \alpha + \beta \in M_1$.

- (i) *There is a unique smooth subgroup H_{M_1} of G , with connected fibers, containing $\underline{\text{Centr}}_G(Q)$ and whose Lie algebra is $\coprod_{\alpha \in M_1} \mathfrak{g}^\alpha$.*
- (ii) *One has the following implications:*

$$\begin{aligned} M_1 = \{0\} &\implies H_{M_1} = \underline{\text{Centr}}_G(Q), \\ M_1 = -M_1 &\implies H_{M_1} \text{ is reductive,} \\ M_1 \cup (-M_1) = M &\implies H_{M_1} \text{ and } H_{-M_1} \text{ are opposite} \\ &\quad \text{parabolic subgroups of } G, \text{ with common} \\ &\quad \text{Levi subgroup } H_{M_1 \cap -M_1}. \end{aligned}$$

To prove (i) and (ii), which are local for the (fpqc) topology, one can assume that Q is contained in a maximal torus T of G ; one can moreover split G relative to T . Assertion (i) then follows at once from Exp. XXII, 5.3.5, 5.4.5 and 5.4.7; the assertions of (ii) follow from Exp. XXII, 5.3.5, 5.10.1, 5.11.3 and from this exposé, 1.4 and 4.3.2.

Corollary 6.2. *Let S be a scheme, G a reductive S -group, Q a split subtorus of G . There is a parabolic subgroup of G of which $\underline{\text{Centr}}_G(Q)$ is a Levi subgroup.*

Indeed, writing $Q = D_S(M)$, one chooses a total ordering on the group M and calls M_1 the set of positive elements of M ; the group H_{M_1} answers the question.

Corollary 6.3. *If the reductive S -group G has a noncentral split subtorus, it has a proper parabolic subgroup (i.e. $\neq G$).*

By 5.9 and 5.10, one deduces from 6.2:

⁽²⁸⁾N.D.E. See [Gi71], § III.3.

Corollary 6.4. *Let S be a scheme, G a reductive S -group, Q a split subtorus of G . The canonical morphism $G \rightarrow G/\underline{\text{Centr}}_G(Q)$ is a locally trivial fibration.*

If S is semi-local, the map $G(S) \rightarrow (G/\underline{\text{Centr}}_G(Q))(S)$ is surjective, and the map $H^1(S, \underline{\text{Centr}}_G(Q)) \rightarrow H^1(S, G)$ is injective.

6.5. — Suppose that S is connected. If T is an S -torus and if T' and T'' are two split subtori of T , their product $T' \cdot T''^{(29)}$ is also a split subtorus of T . Indeed, it is identified with the quotient of $T' \times_S T''$ by $T' \cap T''$, a quotient that is split by Exp. IX, 2.11. It follows that T has a largest split subtorus; one denotes it by T_{spl} .

Lemma 6.6. *Let S be a connected scheme, T an isotrivial S -torus, T_{spl} its largest split subtorus. The following conditions are equivalent:*

- (i) *There is a homomorphism $T \rightarrow \mathbf{G}_{m,S}$ distinct from e .*
- (ii) *$T_{\text{spl}} \neq e$.*

Since T is assumed to be isotrivial, there are a finite group Γ , a connected Galois principal covering $S' \rightarrow S$ with group Γ , and an isomorphism $T_{S'} \simeq D_{S'}(M)$; M is then endowed with a Γ -module structure, and one has a natural isomorphism $\text{Hom}_S(T, \mathbf{G}_{m,S}) = H^0(\Gamma, M)$.

Moreover, let V be the vector subspace of $M \otimes \mathbf{Q}$ generated by the elements of the form $g(m) - m$, $g \in \Gamma$, $m \in M$. One checks at once that $(T_{\text{spl}})_{S'}$ is identified with $D_{S'}(M/M \cap V)$. Assertion (i) is therefore equivalent to $H^0(\Gamma, M) \neq 0$, or again to $H^0(\Gamma, M \otimes \mathbf{Q}) \neq 0$, whereas assertion (ii) is equivalent to $M \neq M \cap V$, or again to $M \otimes \mathbf{Q} \neq V$. Now one has $M \otimes \mathbf{Q} = H^0(\Gamma, M \otimes \mathbf{Q}) \oplus V$, as one checks at once (consider the projector $M \otimes \mathbf{Q} \rightarrow H^0(\Gamma, M \otimes \mathbf{Q})$ that sends x to the average of its transforms by Γ).

Lemma 6.7. *Let S be a scheme, G a reductive S -group, P a parabolic subgroup of G such that $P \neq G$, L a Levi subgroup of P , Q its radical. There is a homomorphism $Q \rightarrow \mathbf{G}_{m,S}$ distinct from e .*

Consider the unipotent radical U of P ; it is invariant under $\text{int}(P)$, hence under $\text{int}(Q)$. Consider the invertible $\mathcal{O}_S$ -module $\det(\text{Lie}(U))$, the “maximum exterior power” of the locally free $\mathcal{O}_S$ -module $\text{Lie}(U)$. The adjoint representation defines a group homomorphism

$$f : Q \rightarrow \underline{\text{Aut}}(\det(\text{Lie}(U))) = \mathbf{G}_{m,S}.$$

If $P \neq G$, then $U \neq e$. Choose an $s \in S$ such that $U_{\bar{s}} \neq e$. Splitting $G_{\bar{s}}$ relative to a maximal torus containing $Q_{\bar{s}}$, one sees at once that $f_{\bar{s}} \neq e$.

Proposition 6.8. *Let S be a connected semi-local scheme, G a reductive S -group, P a parabolic subgroup of G , L a Levi subgroup of P , Q its radical, Q_{spl} the largest split subtorus of Q (i.e. the largest split central subtorus of L). Then*

$$L = \underline{\text{Centr}}_G(Q_{\text{spl}}).$$

Put $L' = \underline{\text{Centr}}_G(Q_{\text{spl}})$; it is a reductive subgroup of G containing L ; moreover, $P' = P \cap L'$ is a parabolic subgroup of L' , with Levi subgroup L (1.20). If $L' \neq L$, then $L' \not\subset P$ (since L is a maximal reductive subgroup of P , cf. 1.7), hence $P' \neq L'$.

Let G_1 be the derived group of G , and $P_1 = P' \cap G_1$. By 1.19, P_1 is a parabolic subgroup of the semisimple group G_1 , $L_1 = L \cap G_1$ is a Levi subgroup of it, and

$$Q_1 = \text{rad}(L_1) = (\text{rad}(L) \cap G_1)^0 = (Q \cap G_1)^0.$$

Since L_1 has a maximal torus T_1 (Exp. XIV, 3.20), and the latter is isotrivial (Exp. XXIV, 4.1.5), Q_1 , which is a subtorus of T_1 , is also isotrivial (Exp. IX, 2.11); since $P_1 \neq G_1$, one can apply 6.7 and 6.6, and therefore has $(Q_1)_{\text{spl}} \neq e$, whence $(Q_{\text{spl}} \cap G_1)^0 \neq e$, hence $Q_{\text{spl}} \not\subset \text{rad}(L')$ (since $\text{rad}(L') \cap G_1$ is finite), which contradicts the definition of L' .

⁽²⁹⁾**N.D.E.** One has adopted multiplicative notation, i.e. replaced “their sum $T' + T''$ ” by “their product $T' \cdot T''$ ”.

Corollary 6.9. *Let S be a connected semi-local scheme, G a reductive S -group, Q a critical subtorus of G (i.e. such that $\text{rad}(\text{Centr}_G(Q)) = Q$). For $\text{Centr}_G(Q)$ to be a Levi subgroup of a parabolic subgroup of G , it is necessary and sufficient that $\text{Centr}_G(Q) = \text{Centr}_G(Q_{\text{spl}})$, that is, that $\text{Lie}(G)^Q = \text{Lie}(G)^{Q_{\text{spl}}}$.*

This follows from 6.2 and 6.8.

Corollary 6.10. *Let S be a connected semi-local scheme, G a reductive S -group, L a subgroup of G . The following conditions are equivalent:*

- (i) *There is a parabolic subgroup of G of which L is a Levi subgroup.*
- (ii) *There is a split subtorus of G of which L is the centralizer.*
- (iii) *There is a homomorphism $\mathbf{G}_{m,S} \rightarrow G$ of which L is the centralizer.*

Indeed, one has (i) $\Rightarrow$ (ii) by 6.8, and (iii) $\Rightarrow$ (i) by 6.1; it remains to prove (ii) $\Rightarrow$ (iii). Suppose therefore that $L = \text{Centr}_G(Q)$, with $Q = D_S(M)$; write $\mathfrak{g} = \text{Lie}(G/S)$ and

$$\mathfrak{g} = \coprod_{\alpha \in M} \mathfrak{g}^\alpha,$$

and let R be the set of the $\alpha \in M - \{0\}$ such that $\mathfrak{g}^\alpha \neq 0$.

Since R is finite and does not contain 0, there is a homomorphism $u : M \rightarrow \mathbf{Z}$ such that $u(\alpha) \neq 0$ for every $\alpha \in R$. By duality, u gives a homomorphism $\mathbf{G}_{m,S} \rightarrow Q$, hence a homomorphism $f : \mathbf{G}_{m,S} \rightarrow G$. One has $\text{Centr}_G(f) \supset \text{Centr}_G(Q)$; these are two smooth subgroups of G , with connected fibers; their Lie algebras coincide (since both are equal to $\mathfrak{g}^0$); they therefore coincide, by a usual argument.

Corollary 6.11. *Let S be a connected semi-local scheme, G a reductive S -group. The maps*

$$L \mapsto \text{rad}(L)_{\text{spl}}, \quad Q \mapsto \text{Centr}_G(Q)$$

are mutually inverse bijections, reversing the natural order structures, between the set of subgroups L of G that are Levi subgroups of parabolic subgroups of G and the set of split subtori Q of G such that $\text{rad}(\text{Centr}_G(Q))_{\text{spl}} = Q$.

Corollary 6.12. *Let S be a connected semi-local scheme, G a reductive S -group. Consider the following assertions:*

- (i) *There is a parabolic subgroup of G distinct from G .*
- (ii) *G has a noncentral split subtorus.*
- (ii bis) *G has a noncentral split subtorus of relative dimension 1.*
- (iii) *There is a group homomorphism $\mathbf{G}_{a,S} \rightarrow G$ that is a closed immersion.*

Then one has (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (ii bis) $\Rightarrow$ (iii).

The only new assertion is (i) $\Rightarrow$ (iii). Let therefore P be a parabolic subgroup of G distinct from G . Then $U = \text{rad}^u(P) \neq e$. Consider the last nontrivial subgroup U_n of the composition series of U (2.1). One has an isomorphism $U_n \simeq W(\mathcal{E}_n)$, where $\mathcal{E}_n$ is a locally free $\mathcal{O}_S$ -module, hence free⁽³⁰⁾. Since $\mathcal{E}_n \neq 0$, there is a monomorphism $\mathcal{O}_S \rightarrow \mathcal{E}_n$ that is locally a direct summand, hence a closed immersion $\mathbf{G}_{a,S} = W(\mathcal{O}_S) \hookrightarrow W(\mathcal{E}_n) \simeq U_n$, which implies (iii) at once.

Remark 6.12.1. When S is the spectrum of a field of characteristic 0, it follows from the Jacobson–Morozov theorem that (iii) $\Rightarrow$ (ii bis). The four preceding conditions are then equivalent (“Godement’s criterion”, cf. [BT65], 8.5).*

⁽³⁰⁾**N.D.E.** Since S is assumed to be semi-local and connected.

*This is more generally true when S is the spectrum of a perfect field (Tits).³¹

⁽³¹⁾**N.D.E.** This is Corollary 3.7 of the article [BT71].

Definition 6.13. Let S be a connected semi-local scheme, G a reductive S -group. One says that G is *anisotropic* if G contains no split subtorus not reduced to e .

Corollary 6.14. Let S be a connected semi-local scheme. For the reductive S -group G to be anisotropic, it is necessary and sufficient that it have no parabolic subgroup $P \neq G$ and that its radical be anisotropic.

Using now 6.6, Exp. XXIV, 4.1.5, and Exp. XXII, 6.2, one deduces:

Corollary 6.15. Let S be a connected semi-local scheme, G an isotrivial reductive S -group (for example, G semisimple, or S normal (Exp. X 5.16)). For G to be anisotropic, it is necessary and sufficient that G have no parabolic subgroup $P \neq G$ and that $\text{Hom}_{S\text{-gr.}}(G, \mathbf{G}_{m,S}) = e$.

Proposition 6.16. Let S be a connected semi-local scheme, G a reductive S -group. The maximal split subtori of G are the largest split central subtori of the Levi groups of the minimal parabolic subgroups of G . Two such tori are conjugate by an element of $G(S)$.

Let Q be a maximal split subtorus of G .⁽³²⁾ Then, by 6.2, $L = \text{Centr}_G(Q)$ is a Levi subgroup of a parabolic subgroup P of G , and since $Q \subset \text{rad}(\text{Centr}_G(Q))_{\text{spl}}$, the maximality of Q implies $Q = \text{rad}(\text{Centr}_G(Q))_{\text{spl}}$. By 6.11, L is a minimal element of the set of Levi subgroups of parabolic subgroups of G , hence P is a minimal parabolic subgroup of G by 1.20. It then follows from 5.7 and 5.5(iv) that two subtori such as Q are conjugate by a section of $G(S)$. The conjugacy of the Q and of the pairs (P, L) then implies the first assertion of 6.16.

Corollary 6.17. Let S be a connected semi-local scheme, P and P' two minimal parabolic subgroups in standard position (4.5.1.1). Then $P \cap P'$ contains a common Levi subgroup of P and P' .

Indeed, $P \cap P'$ contains a maximal torus T of G , by 4.5.1(v); let L be the unique Levi subgroup of P containing T . One has

$$\text{rad}(P) \cap T = \text{rad}(P) \cap L = \text{rad}(L)$$

by 1.21, hence $\text{rad}(P) \cap T$ contains $\text{rad}(L)_{\text{spl}}$, which is a maximal split subtorus of G , and is therefore necessarily equal to T_{spl} . One therefore has $L = \text{Centr}_G(T_{\text{spl}})$, and by symmetry L is also a Levi subgroup of P' .

Remark 6.18. It follows from 1.21 that the parabolic subgroup P of G is minimal if and only if $\text{rad}(P)$ contains a maximal split subtorus of G ; then, by the proof of 6.17, if T is a maximal torus of P , T_{spl} is a maximal split torus of G and of $\text{rad}(P)$, and $\text{Centr}_G(T_{\text{spl}})$ is a Levi subgroup of P . Moreover, every Levi subgroup of P is obtained in this way.

7. Relative root datum

In this paragraph, S will denote a *nonempty connected semi-local* scheme, G a reductive S -group, Q a maximal split subtorus of G , and L the centralizer of Q in G , i.e. $L = \text{Centr}_G(Q)$.

7.1. — Since Q is the largest split central subtorus of L , every section of $G(S)$ that normalizes L also normalizes Q . One therefore has (cf. 7.1.1)

$$\text{Norm}_G(L)(S) = \text{Norm}_G(Q)(S).$$

Moreover, one has seen in 6.4 that the map $G(S) \rightarrow (G/L)(S)$ is surjective. It follows that one has a canonical identification

$$W_G(Q)(S) = (\text{Norm}_G(Q)/\text{Centr}_G(Q))(S) \simeq \text{Norm}_G(L)(S)/L(S).$$

One will denote by M the group $\text{Hom}_{S\text{-gr.}}(Q, \mathbf{G}_{m,S})$, so that one has a canonical isomorphism $Q \simeq D_S(M)$. One will write W for the group of automorphisms of M defined by $W_G(Q)(S)$.

⁽³²⁾**N.D.E.** One has expanded the sentence that follows.

One therefore has isomorphisms

$$W \simeq W_G(Q)(S) \simeq \underline{\text{Norm}}_G(L)(S)/L(S).$$

7.1.1. — In general, one does not have $\underline{\text{Norm}}_G(L) = \underline{\text{Norm}}_G(Q)$. Take, for example, for S the spectrum of a field k having a quadratic extension k' , and for G the unitary group $\text{Q-}\hat{\text{E}}_{\text{p}_{k'}/k}(A_2)$ (cf. Exp. XXIV, 3.11.2).

Since the minimal parabolic subgroups of G are its Borel subgroups, their Levi subgroups are maximal tori, and one has $(\underline{\text{Norm}}_G(L)/L)(\bar{k}) = \mathfrak{S}_3$.

Moreover, since G is not split, the maximal split tori of G are of dimension ≤ 1 , hence isomorphic to $\mathbf{G}_{m,k}$. Since $\underline{\text{Norm}}_G(Q)/L$ acts faithfully on Q , one has $(\underline{\text{Norm}}_G(Q)/Q)(\bar{k}) = \mathbf{Z}/2\mathbf{Z}$.

7.2. — If P is a parabolic subgroup of G with Levi group L (such a subgroup exists by 6.2), P is necessarily minimal (cf. 6.18). By the conjugacy of minimal parabolic subgroups of G (5.7), that of Levi subgroups of a parabolic subgroup (1.8), and the equalities $P = \underline{\text{Norm}}_G(P)$ and $\underline{\text{Norm}}_G(L) \cap P = L$ (1.6), one obtains: *the set of (minimal) parabolic subgroups of G with Levi group L is principal homogeneous under the group W .*

7.3. — The Lie algebra of G decomposes under the action of Q as

$$\text{Lie}(G) = \text{Lie}(L) \oplus \coprod_{\alpha \in R} \text{Lie}(G)^\alpha,$$

where R is the set of nonzero characters of Q such that $\text{Lie}(G)^\alpha \neq 0$ (roots of G relative to Q).

Denote by M^* the group $\text{Hom}_{S\text{-gr.}}(\mathbf{G}_{m,S}, Q)$, which is in duality with M and on which W acts naturally by transport of structure.

Theorem 7.4. *With the notation of 7.3, there is a unique map $\alpha \mapsto \alpha^*$ from R to M^* defining in (M, M^*) a root datum (Exp. XXI, 1.1) whose Weyl group is W .*

Moreover, the parabolic subgroups P of G with Levi group L and the systems of positive roots R_+ of R correspond bijectively by the relation

$$\text{Lie}(P) = \text{Lie}(L) \oplus \coprod_{\alpha \in R_+} \text{Lie}(G)^\alpha.$$

7.4.1. — Suppose first that the existence of the desired map $\alpha \mapsto \alpha^*$ has been proved. By Exp. XXI, 3.4.10, s_α is the unique element of W such that for every $m \in M$, $s_\alpha(m) - m$ is a rational multiple of α , which shows that s_α is determined by α ; since one then has⁽³³⁾ $\alpha^*(m)\alpha = m - s_\alpha(m)$, one sees that α^* is determined by α , which proves the uniqueness of the map $\alpha \mapsto \alpha^*$.

7.4.2. — Let $\alpha \in R$, and let L_α (resp. H_α , resp. $H_{-\alpha}$) be the unique smooth subgroup with connected fibers of G containing L and such that (cf. 6.1(i))

$$\text{Lie}(L_\alpha) = \text{Lie}(L) \oplus \coprod_{\gamma \in \mathbf{Z}\alpha \cap R} \text{Lie}(G)^\gamma,$$

resp.

$$\text{Lie}(H_\alpha) = \text{Lie}(L) \oplus \coprod_{\gamma \in \mathbf{N}\alpha \cap R} \text{Lie}(G)^\gamma,$$

resp.

$$\text{Lie}(H_{-\alpha}) = \text{Lie}(L) \oplus \coprod_{\gamma \in -\mathbf{N}\alpha \cap R} \text{Lie}(G)^\gamma;$$

L_α is a reductive subgroup of G , H_α and $H_{-\alpha}$ are parabolic subgroups of it with Levi subgroup L , and H_α and $H_{-\alpha}$ are opposite relative to L (cf. 6.1(ii)).

⁽³³⁾**N.D.E.** One has corrected $\alpha^*(m)$ to $\alpha^*(m)\alpha$.

By 7.2, there is therefore $s_\alpha \in \underline{\text{Norm}}_{L_\alpha}(Q)(S)/L(S) \subset W$ such that $s_\alpha(H_\alpha) = H_{-\alpha}$. One has $s_\alpha(\alpha) = -\alpha$ (since α (resp. $-\alpha$) is the common divisor of the elements of R occurring in H_α (resp. $H_{-\alpha}$)), and one has $s_\alpha^2 = \text{id}$ (since $s_\alpha^2(H_\alpha)$ and H_α are both opposite to $H_{-\alpha}$ relative to L). One has therefore constructed an $s_\alpha \in W$ satisfying the following properties:

$$(x) \quad s_\alpha(\alpha) = -\alpha, \quad s_\alpha^2 = \text{id},$$

$$(xx) \quad s_\alpha \text{ can be represented by an element of } L_\alpha(S).$$

Observe, moreover, that s_α is constructed canonically from α , and in particular that

$$(xxx) \quad \text{for every } w \in W, \text{ one has } ws_\alpha w^{-1} = s_{w(\alpha)}.$$

7.4.3. — We now propose to prove the assertion:

$$(xxxx) \quad \text{for every } m \in M, \quad s_\alpha(m) - m \in \mathbf{Z}\alpha.$$

Since S is connected, this assertion is local for the (fpqc) topology. One can therefore assume that $L_\alpha = G_1$ is splittable relative to a maximal torus T_1 of L . Let therefore (G_1, T_1, M_1, R_1) be such a splitting. The monomorphism $Q \rightarrow T_1$ identifies M with a quotient of M_1 ; let $p: M_1 \rightarrow M$ be the canonical map.

The image of R_1 under p consists possibly of zero and of the roots of G_1 relative to Q (hence of the elements of R that are integral multiples of α); one therefore has $p(R_1) \subset \mathbf{Z}\alpha$. By (xx), there is an element of $\underline{\text{Norm}}_{G_1}(L)(S)$ that induces s_α on Q . By Exp. XXII, 5.10.10, there is therefore a section $w \in (W_1)_S(S)$ that induces s_α on Q (one denotes by W_1 the Weyl group of the root datum $(M_1, R_1, \dots)$). After restricting S , one can therefore assume that there is $w \in W_1$ inducing s_α on Q , hence satisfying $p(w(m_1)) = s_\alpha(p(m_1))$ for every $m_1 \in M_1$. But, by definition of W_1 , w is a product of reflections with respect to elements of R_1 , hence $w(m_1) - m_1$ is a linear combination with integer coefficients of the elements of R_1 . It follows that $s_\alpha(p(m_1)) - p(m_1)$ is a linear combination with integer coefficients of the elements of $p(R_1) \subset \mathbf{Z}\alpha$, hence an integral multiple of α , which was to be proved.

7.4.4. — One can therefore define an element $\alpha^* \in M^*$ by⁽³⁴⁾

$$\alpha^*(m)\alpha = m - s_\alpha(m).$$

By (x), one has $(\alpha^*, \alpha) = 2$; moreover, it follows from (xxx) that for every pair $(\alpha, \alpha') \in R \times R$, one has $s_\alpha(\alpha') \in R$ and $s_\alpha(\alpha^*) = s_\alpha(\alpha')^*$, which proves (cf. Exp. XXI, 1.1) that the map $\alpha \mapsto \alpha^*$ constructed does indeed define a root datum in (M, M^*) .

7.4.5. — Let W' be the Weyl group of this root datum (the group of transformations of M generated by the s_α); one has $W' \subset W$.

Let, moreover, $\geq$ be a total ordering on the free abelian group M ; put $R_+ = \{\alpha \in R \mid \alpha \geq 0\}$. One knows that R_+ is a system of positive roots of R . Let $w \in W$, represented by an $n \in \underline{\text{Norm}}_G(L)(S) = \underline{\text{Norm}}_G(Q)(S)$. Put $P = H_{R_+}$ (notation of 6.1); by *loc. cit.*, P is a parabolic subgroup of G , with Levi subgroup L . One obviously has $\text{int}(n)P = H_{w(R_+)}$. It then follows from 7.3 that $w(R_+) = R_+$ implies $w = e$. Since the group W' acts transitively on the systems of positive roots of R (Exp. XXI, 3.3.7), and since the stabilizer of R_+ in W is the identity, one concludes at once that $W = W'$. One also concludes that $W = W'$ acts simply transitively both on the set of systems of positive roots of R and on the set of parabolic subgroups of G with Levi group L , which implies the last assertion of 7.4. Q.E.D.

7.5. — If P and P_1 are two minimal parabolic subgroups of G , with Levi subgroups L and L_1 , and if one denotes by Q and Q_1 the maximal split central tori of L and L_1 , then the pairs (P, Q) and (P_1, Q_1) are conjugate: there is $g \in G(S)$ such that $\text{int}(g)P = P_1$ and $\text{int}(g)Q = Q_1$.

⁽³⁴⁾**N.D.E.** One has corrected $\alpha^*(m)$ to $\alpha^*(m)\alpha$.

Indeed, P and P_1 are conjugate (5.7), and one can therefore assume that $P = P_1$; then L and L_1 are conjugate by a section of $P(S)$ (1.8). Moreover, if g and g' are two sections of G conjugating the pairs (P, Q) and (P_1, Q_1) , then $g'^{-1}g$ normalizes P and Q , hence P and L ; but

$$\underline{\text{Norm}}_G(P) \cap \underline{\text{Norm}}_G(L) = P \cap \underline{\text{Norm}}_G(L) = L = \underline{\text{Centr}}_G(Q).$$

The isomorphism $Q \xrightarrow{\sim} Q_1$ induced by $\text{int}(g)$ is therefore independent of g .

Let $\mathcal{R}$ and $\mathcal{R}_1$ be the root data defined by means of 7.4 in $\text{Hom}_{S\text{-gr.}}(Q, \mathbf{G}_{m,S})$ and $\text{Hom}_{S\text{-gr.}}(Q_1, \mathbf{G}_{m,S})$, and let R_+ and R_{1+} be the systems of positive roots corresponding to P and P_1 . The canonical isomorphism $Q \xrightarrow{\sim} Q_1$ defined above transforms $(\mathcal{R}, R_+)$ into $(\mathcal{R}_1, R_{1+})$. One deduces at once that one can define the *pinned relative root datum*⁽³⁵⁾ of G over S , by identifying the different $(\mathcal{R}, R_+)$ by means of the transitive system of isomorphisms described above.

From now on, we shall denote this pinned root datum by

$$(\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}_+) = \mathcal{R}(G/S);$$

for every pair $P \supset Q$ as above, one therefore has a canonical isomorphism

$$\underline{M} \xrightarrow{\sim} \text{Hom}_{S\text{-gr.}}(Q, \mathbf{G}_{m,S})$$

transforming $\underline{R}$ (resp. $\underline{R}_+$) into the set of roots of G (resp. of P) relative to Q , and $W(\underline{R})$ into $W_G(Q)(S)$.

7.6. — Let Q still be a maximal split torus of G , P a (minimal) parabolic subgroup of G with Levi group $\underline{\text{Centr}}_G(Q)$, $(\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}_+)$ the corresponding pinned root datum (7.4), and Δ the set of simple roots of R_+ . For every $A \subset \Delta$, let $R_A \subset R$ be the set

$$R_A = R_+ \cup (\mathbf{Z}A \cap R_-)$$

consisting of the positive roots and the negative roots that are linear combinations of the elements of A . It is a closed set (Exp. XXI, 3.1.4) of roots, and every closed set containing R_+ can be written uniquely in this form (Exp. XXI, 3.3.10). By 6.1, there is a unique subgroup P_A of G , smooth and with connected fibers, containing $\underline{\text{Centr}}_G(Q)$ and such that

$$\text{Lie}(P_A) = \text{Lie}(G)^0 \oplus \coprod_{\alpha \in R_A} \text{Lie}(G)^\alpha.$$

It then follows at once from 6.1, from the conjugacy of minimal parabolic subgroups, and from the fact that the set of roots of a parabolic subgroup of G containing Q is closed (which follows at once from 1.4 by splitting), that:

Proposition 7.7.

- (i) *The map $A \mapsto P_A$ is a bijection from the set of subsets of Δ onto the set of parabolic subgroups of G containing P . This bijection preserves the natural inclusion order relations.*
- (ii) *Every parabolic subgroup of G is conjugate by a section of $G(S)$ to a unique P_A .*

7.8. — Let $P \supset Q$ be as above. Consider the relative root datum (7.5) of G over S and the canonical isomorphism

$$f : (\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}_+) \xrightarrow{\sim} (\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}_+).$$

The set Δ of simple roots of R_+ is transformed into the set $\underline{\Delta}$ of simple roots of $\underline{R}_+$, hence every subset A of Δ into a subset $f(A) \subset \underline{\Delta}$.

⁽³⁵⁾**N.D.E.** Recall (Exp. XXIII 1.5) that a pinned root datum is a root datum endowed with a choice of a system of positive roots (or of simple roots).

Definition 7.9.0. ⁽³⁶⁾Let H be any parabolic subgroup of G . By 7.7(ii), it is conjugate to a unique P_A . Write $t_r(H) = f(A) \subset \underline{\Delta}$. One checks at once, by means of the conjugacy theorems, that $t_r(H)$ is independent of the choice of the pair $P \supset Q$. One says that it is the *relative type* of H .

Proposition 7.9.

- (i) The map $H \mapsto t_r(H)$ induces a bijection between the set of conjugacy classes (by $G(S)$) of parabolic subgroups of G and the set of subsets of $\underline{\Delta}$.
- (ii) Let H be a parabolic subgroup of G , P a minimal parabolic subgroup contained in H , Q the maximal split central torus of a Levi subgroup of P , Δ the set of simple roots of P relative to Q , and $f : \Delta \xrightarrow{\sim} \underline{\Delta}$ the canonical isomorphism. Then, for every $\alpha \in \Delta$, one has the equivalence:

$$f(\alpha) \in t_r(H) \iff \text{Lie}(H)^{-\alpha} \neq 0,$$

and one has $H = P_A$, where $A = f^{-1}(t_r(H))$.

- (iii) If H and H' are two parabolic subgroups of G containing P , then (see 3.8.1(ii) and 5.5(i) for other equivalent conditions):

$$t_r(H) \subset t_r(H') \iff t(H) \subset t(H').$$

7.10. — One can study the relative positions of two minimal parabolic subgroups; the results are as follows (one refers to 4.5.2 for the notation $t_2(P, P_1)$):

- (1) If P, P_1, P', P'_1 are four minimal parabolic subgroups of G , then $t_2(P, P_1) = t_2(P', P'_1)$ (i.e. (P, P_1) and (P', P'_1) are conjugate locally for (fpqc)) if and only if there is $g \in G(S)$ such that $\text{int}(g)P = P'$ and $\text{int}(g)P_1 = P'_1$.
- (2) Fix, in particular, a minimal parabolic subgroup P with Levi subgroup L , and let T be a maximal torus of L . Consider the scheme $\underline{\text{Par}}_{t_{\min}}(G; P)$ of minimal parabolic subgroups of G in standard position relative to P . One has a morphism (cf. 4.5.5)

$$f : \underline{\text{Par}}_{t_{\min}}(G; P) \longrightarrow W_P(T) \backslash W_G(T) / W_P(T),$$

whose fibers are “the orbits of P in $\underline{\text{Par}}_{t_{\min}}(G; P)$ ”. By (1), f therefore induces a monomorphism

$$P(S) \backslash \underline{\text{Par}}_{t_{\min}}(G; P)(S) \hookrightarrow (W_P(T) \backslash W_G(T) / W_P(T))(S).$$

The image of this morphism is identified with W ; this is the *Bruhat theorem*: every orbit of $P(S)$ in $\underline{\text{Par}}_{t_{\min}}(G; P)(S)$ contains one and only one parabolic subgroup of G with Levi group L (that is, of the form $\text{int}(n)P$, where $n \in \underline{\text{Norm}}_G(L)$).

- (3) In other words, let $E(S)$ be the set of $g \in G(S)$ such that $\text{int}(g)P$ and P are in mutual standard position. Then $E(S)$ has a partition into double cosets modulo $P(S)$ indexed by W :

$$E(S) = P(S)W P(S)$$

(obvious notation). Putting $U = \text{rad}^u(P)(S)$, one can also write

$$E(S) = U(S) \cdot \underline{\text{Norm}}_G(L)(S) \cdot U(S),$$

thus exhibiting a partition of $E(S)$ into double cosets modulo $U(S)$, indexed by $\underline{\text{Norm}}_G(L)(S)$.

- (4) If S is the spectrum of a field, then $E(S) = G(S)$, and one recovers [BT65], 5.15.

Counterexamples 7.11. Let $S = \text{Spec}(\mathbf{Z}/4\mathbf{Z})$, $G = \text{SL}_{2,S}$. Let B be the usual Borel subgroup consisting of the matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $c = 0$. Let $g = \begin{pmatrix} -1 & 1 \\ 2 & 1 \end{pmatrix} \in G(S)$, and put $B' = \text{int}(g)B$. Then $B(S) = B'(S)$, and $B \cap B'$ contains no maximal torus⁽³⁷⁾. This shows,

⁽³⁶⁾**N.D.E.** One has added the numbering 7.9.0 to bring out the definition of “relative type”.

⁽³⁷⁾**N.D.E.** Indeed, B' is the subgroup of G defined by the equation $c = 2(a + b)$; then $B \cap B'$ is not flat over S , hence contains no maximal torus, by 4.5.1.

on the one hand, that two distinct minimal parabolic subgroups can have the same group of sections; on the other hand, that in general there is no criterion allowing one to recognize whether two minimal parabolic subgroups P and P' are in standard position solely by means of the groups $P(S)$ and $P'(S)$. In particular, the subset $E(S)$ of $G(S)$ does not seem to be definable solely by means of the situation $\{G(S), P(S), \underline{\text{Norm}}_G(L)(S)\}$ (in the preceding case, this subset is defined by $c \neq 2^{(38)}$).

7.12. — One now proposes to study the variation of $\mathcal{R}(G/S)$ with S . Let therefore S' be an S -scheme, also semi-local, connected and nonempty. Let Q be a maximal split torus of G ; then $Q_{S'}$ is a split torus of $G_{S'}$; let Q' be a maximal split torus of $G_{S'}$ containing $Q_{S'}$. Put

$$M = \text{Hom}_{S\text{-gr.}}(Q, \mathbf{G}_{m,S}) \simeq \text{Hom}_{S'\text{-gr.}}(Q_{S'}, \mathbf{G}_{m,S'}), \\ M' = \text{Hom}_{S'\text{-gr.}}(Q', \mathbf{G}_{m,S'}).$$

The monomorphism $Q_{S'} \rightarrow Q'$ induces an epimorphism $u : M' \rightarrow M$. Write

$$L = \underline{\text{Centr}}_G(Q), \quad L' = \underline{\text{Centr}}_{G_{S'}}(Q');$$

one has $L' \subset L_{S'}$.

If H is a subgroup of G containing L , then $H_{S'}$ contains L' , and one has

$$\text{Lie}(H) = \text{Lie}(L) \oplus \coprod_{\alpha \in R_H} \text{Lie}(H)^\alpha, \\ \text{Lie}(H_{S'}) = \text{Lie}(L') \oplus \coprod_{\alpha' \in R'_{H_{S'}}} \text{Lie}(H_{S'})^{\alpha'},$$

where R_H (resp. $R'_{H_{S'}}$) denotes the set of roots of H (resp. $H_{S'}$) relative to Q (resp. Q'). One immediately deduces that

$$R_H \subset u(R'_{H_{S'}}) \subset R_H \cup \{0\}.$$

Taking $H = G$, one first sees that $R \subset u(R') \subset R \cup \{0\}$; then taking for H a minimal parabolic subgroup P with Levi subgroup L , one sees that $R'_{H_{S'}}$ contains a system of positive roots of R' , hence (7.4) that there is a minimal parabolic subgroup P' of $G_{S'}$ with Levi subgroup L' contained in $P_{S'}$. One has therefore constructed a diagram

$$\begin{array}{ccccc} Q_{S'} & \subset & L_{S'} & \subset & P_{S'} \\ \cap & & \cup & & \cup \\ Q' & \subset & L' & \subset & P'. \end{array}$$

If R_+ (resp. Δ) is the system of positive (resp. simple) roots of R defined by P , and if one defines R'_+ and Δ' similarly, one readily checks that

$$R_+ \subset u(R'_+) \subset R_+ \cup \{0\}, \quad \Delta \subset u(\Delta') \subset \Delta \cup \{0\}.$$

Let now $w \in W \simeq \underline{\text{Norm}}_G(Q)(S)/\underline{\text{Centr}}_G(Q)(S)$, and let $n \in \underline{\text{Norm}}_G(Q)(S)$ be a representative of w . One has $\text{int}(n)Q = Q$, hence $\text{int}(n)L = L$, hence $\text{int}(n)L_{S'} = L_{S'}$. Then Q' and $\text{int}(n)Q'$ are two maximal split tori of $L_{S'}$, hence are conjugate by a section $x \in L(S')$, and one has $\text{int}(nx)Q' = Q'$, hence $nx \in \underline{\text{Norm}}_{G_{S'}}(Q')(S')$. Let w' be the image of $n' = nx$ in $W' \simeq \underline{\text{Norm}}_{G_{S'}}(Q')(S')/\underline{\text{Centr}}_{G_{S'}}(Q')(S')$. It is clear that the action of w' on M' is compatible with the projection $u : M' \rightarrow M$, and that the induced action on M coincides with the one defined by w .

Using now the definition of relative root data and the conjugacy theorems, one proves without difficulty:

⁽³⁸⁾**N.D.E.** More generally, for every S -scheme S' , $E(S')$ is defined by the condition: “ c is zero or invertible”.

Theorem 7.13. *Let S and S' be two nonempty connected semi-local schemes, $S' \rightarrow S$ a morphism of S -schemes, G a reductive S -group,*

$$\begin{aligned}\mathcal{R}(G/S) &= (\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}_+), \\ \mathcal{R}(G_{S'}/S') &= (\underline{M}', \underline{M}'^*, \underline{R}', \underline{R}'^*, \underline{R}'_+)\end{aligned}$$

the relative pinned root data. There is a canonical homomorphism

$$\underline{u} : \underline{M}' \rightarrow \underline{M}$$

satisfying the following conditions:

- (i) $\underline{u}$ is surjective.
- (ii) For every $w \in \underline{W}$, there is an element w' of $\underline{W}'$ compatible with $\underline{u}$ and inducing w on $\underline{M}$.
- (iii) For every subset X of $\underline{M}$, write $X^\wedge = X \cap (\underline{M} - \{0\})$. Then

$$\underline{u}(\underline{R}'_+)^{\wedge} = \underline{R}_+, \quad \underline{u}(\underline{\Delta}')^{\wedge} = \underline{\Delta}.$$

- (iv) For every parabolic subgroup H , consider $t_r(H) \subset \underline{\Delta}$ and $t_r(H_{S'}) \subset \underline{\Delta}'$. Then

$$\begin{aligned}t_r(H_{S'}) &= \underline{u}^{-1}(t_r(H) \cup \{0\}) \cap \underline{\Delta}' \\ &= \{\alpha' \in \underline{\Delta}' \mid \underline{u}(\alpha') \in t_r(H) \text{ or } \underline{u}(\alpha') = 0\}.\end{aligned}$$

Remark 7.14. If G is splittable over S , its maximal split tori are maximal tori, and the relative notions introduced here then coincide with the absolute notations already introduced. The preceding theorem therefore gives a description of the relative root datum $\mathcal{R}(G/S)$ and of the relative type t_r by means of the absolute root datum and the absolute type of the group $G_{S'}$, where S' is chosen in such a way that $G_{S'}$ is splittable (cf. Exp. XXIV, 4.4.1). Let us refer to [BT65], 6.12 and following, for this description.

7.15. — Let S be a *henselian local* scheme, s_0 its closed point, S_0 the spectrum of the residue field of s_0 , identified with a closed subscheme of S ; for every object X over S , denote by X_0 the object over S_0 obtained from X by base change. Let, finally, G be a reductive S -group. For every parabolic subgroup P of G , P_0 is a parabolic subgroup of G_0 ; conversely, for every parabolic subgroup $\overline{P}$ of G_0 , there is a parabolic subgroup P of G such that $P_0 = \overline{P}$ (this follows from Hensel's lemma and the fact that $\text{Par}(G)$ is a smooth S -scheme); in particular (cf. 5.7), a parabolic subgroup P of G is minimal if and only if P_0 is minimal. Once such a subgroup P of G has been chosen, an analogous argument shows that the maximal split subtori of P_0 are of the form T_0 , where T is a maximal split subtorus of P . It follows without difficulty that *the relative root data of G over S and of G_0 over S_0 are canonically isomorphic*, so that the theory of parabolic subgroups of G reduces to that of parabolic subgroups of G_0 .

Observe, moreover, that every reductive S_0 -group is of the form G_0 (Exp. XXIV, Prop. 1.21), which conversely allows one to reduce the study of parabolic subgroups of a reductive S_0 -group to the corresponding study over S .

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⁽³⁹⁾**N.D.E.** To this reference, which appeared in the original, one has added the references that follow.